

Looped_Transformer

2026 年 4 月 29 日

1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ ，每层的输入是前一层输出与初始输入的和：

$$\begin{aligned} h_0 &= \text{input} \\ h_l &= \text{TransformerBlock}(h_{l-1} + h_0 | \theta) \quad \text{for } l = 1, 2, \dots, L \end{aligned}$$

1.1.1 截断损失函数

$$\text{Loss}(\theta) = \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^b \frac{1}{k+1} \sum_{i=0}^k (Y_t(P^i | \theta) - f(x_{i+1}))^2 \right]$$

其中 - $\min_{\theta}$ ：这是我们在寻找一组最优的物理定律（模型参数 θ ）。 - $\mathbb{E}_P$ ：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求“系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0} \sum_{t=b_0}^b$ ：这是时间平均。 b 是总迭代次数， T 是窗口大小， $b_0 = \max(b - T, 0)$ 。我们只关心系统演化到“稳态”附近的这段时间内（也就是 b_0 到 b 的表现），前面瞬态的误差我们不计较。 - $\frac{1}{k+1} \sum_{i=0}^k$ ：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i | \theta)$ ：在时间 t 、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
 - 维度设定：问题维度 $d = 20$ ，上下文样本数 $k = 40$ 。
 - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$ 。
 - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本） $x_i \sim \mathcal{N}(0, I_d)$ 。

- 计算标签: $y_i = w^T x_i$ 。
- 拼接成 Prompt: 最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \dots, x_k, y_k, x_{test}, y_{dummy})$ 。
- 线性回归可以进一步改成非线性回归 (比如 ReLU 与 Linear 叠加), 或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间 双通道映射与“拉链式”交织

由于 x 和 y 代表不同的物理量 (一个是空间坐标/特征, 一个是目标观测值), 我们需要赋予它们独立的投影空间:

1. 定义两个独立的线性层:

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 k 个 x_i 丢给 `read_in_x`, 变成张量 X_{emb} , 形状为 $[batch, k, 256]$ 。
3. 把前 k 个 y_i 丢给 `read_in_y`, 变成张量 Y_{emb} , 形状为 $[batch, k, 256]$ 。
4. 像拉拉链一样, 在 `seq_len` 维度上把 X_{emb} 、 Y_{emb} 交织 (Interleave) 起来, 拼接成最终长度为 $2k$ 的提示向量 P 。

在工程实现中, 我们在 x_{test} 后填充了一个 y_{dummy} (如 0), 使实际拼接长度变为 $2k+2$ (偶数)。因此, 真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置, 即索引 $[-2]$ 处。

1.2 代码

```
[112]: import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
from operator import sub
import copy
import numpy as np
import math
import concurrent.futures
import threading
```

1.2.1 位置编码 (PE, Position Encoding)

绝对位置编码 (APE, Absolute Position Encoding)

```
[113]: class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) #
        ↪ [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
        # APE 核心: 直接生成一个完整的 pe 矩阵, 偶数维度放 sin, 奇数维度放 cos
        stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)),
        ↪ dim=-1) # [max_seq_len, d_model/2, 2]
        pe=stacked.flatten(1,2) # [max_seq_len, d_model]
        self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len,
        ↪ d_model]

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        return x + self.pe[:, :x.shape[1], :]
```

Learned APE

```
[114]: class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)
```

ALiBi (Attention with Linear Biases) ALiBi 通过更改内积的方式引入线性位置偏置, 从而实现了位置编码的功能。

$$\text{Score}(q_i, k_j) = q_i \cdot k_j - m \cdot (i - j) \quad (i \geq j)$$

其中 m 是一个与头数 (H) 相关的斜率参数, 硬编码为

$$m_h = 2^{-\frac{sh}{H}} \quad h = 0, 1, \dots, H - 1$$

```
[115]: class ALiBi(nn.Module):
    def __init__(self, num_heads, max_seq_len=4096):
        super().__init__()
        dist = torch.arange(max_seq_len) # [max_seq_len]
        dist_matrix = torch.abs(dist.view(-1, 1) - dist.view(1, -1)) #
        ↪ [max_seq_len, max_seq_len]
        bias = -(2**-(8*torch.arange(1,num_heads+1) / num_heads)).view(-1, 1,
        ↪ 1) * dist_matrix.unsqueeze(0) # [num_heads, max_seq_len, max_seq_len]
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
        ↪ tril (triangular lower) 提取下三角元素，并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
        ↪ float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
        self.register_buffer('fused_mask', bias + mask) # [num_heads,
        ↪ max_seq_len, max_seq_len]

    def forward(self, seq_len):
        # score shape: [batch_size, num_heads, seq_len, seq_len]
        return self.fused_mask[:, :seq_len, :seq_len]
```

原来的 RoPE (Rotary Position Encoding)

```
[116]: class RoPE(nn.Module):
    def __init__(self, d_k, max_seq_len = 4096, b=10000):
        super().__init__()
        theta_i = 1/(b**(torch.arange(0,d_k,2).float()/d_k)) # \theta_i =
        ↪ b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        m = torch.arange(max_seq_len).float() # m = [0,1,...,seq_len-1]
        m_theta_i = torch.outer(m, theta_i) # [seq_len, d_k/2]
        cos = torch.cos(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
        ↪ d_k]
        sin = torch.sin(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
        ↪ d_k]
        self.register_buffer('cos', cos[None, None, :, :]) # 好处: cos 和 sin
        不需要更新参数, 注册为 buffer 后会自动放到正确的设备上
        self.register_buffer('sin', sin[None, None, :, :]) # 扩充维度以适应后续
        计算
```

```

def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
    # x: [batch_size, num_heads, seq_len, d_k]
    seq_len = x.shape[2]
    d_2 = x.shape[-1] // 2
    cos = self.cos[:, :, :seq_len, :] # [1, 1, seq_len, d_k]
    sin = self.sin[:, :, :seq_len, :]
    x_first_half = x[..., :d_2]
    x_second_half = x[..., d_2:]
    x_flip = torch.cat((-x_second_half, x_first_half), dim=-1)
    x_out = x * cos + x_flip * sin # 旋转位置编码
    return x_out

```

自创: Multi-Scale Untied Positional Encoding (MS-UPE)

```

[117]: class MS_UPE(nn.Module):
    def __init__(self, num_heads, d_k, max_seq_len = 4096, b_0=10000,
        ↪ head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio*(torch.arange(0, num_heads))))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0, 1, \dots, \frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)*(torch.arange(0, d_k, 2).float()/d_k).
        ↪ unsqueeze(0)) # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float() # m = [0, 1, ..., seq_len-1]
        m_theta_i = torch.einsum('m,hk->hmk', m, theta_i) # [num_heads,
        ↪ seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1) #
        ↪ [num_heads, seq_len, d_k]
        self.register_buffer('pe', pe[None, :, :, :]) # 好处: pe 不需要更新参数,
        注册为 buffer 后会自动放到正确的设备上

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        pe = self.pe[:, :, :seq_len, :] # [1, num_heads, seq_len, d_k]
        x_out = x + pe
        return x_out

```

1.2.2 Multi-Head Attention

注：APE 应该加在 ToyModel 上而不是 TransformerBlock 上，因为我们只需要最开头输入处编码位置，而不是叠加 num_blocks 次。

```
[118]: class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096,
        ↪pe_type='learned_ape', b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by
        ↪num_heads"
        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads
        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)
        # 位置编码模块
        self.pe_type = [pe_type] if isinstance(pe_type, str) else pe_type
        self.qk_pe_modules = nn.ModuleList()
        self.sc_pe_modules = nn.ModuleList()
        for pe in self.pe_type:
            if pe == 'rope':
                pe_module = RoPE(self.d_k, max_seq_len, b=b_rope_or_upe)
                self.qk_pe_modules.append(pe_module) # RoPE 应用于 Q 和 K
            elif pe == 'ms_upe':
                pe_module = MS_UPE(num_heads, self.d_k, max_seq_len,
                ↪b_0=b_rope_or_upe, head_ratio=head_ratio_upe)
                self.qk_pe_modules.append(pe_module) # MS-UPE 应用于 Q 和 K
            elif pe == 'alibi':
                pe_module = ALiBi(num_heads, max_seq_len)
                self.sc_pe_modules.append(pe_module) # ALiBi 应用于 score 矩阵
            elif pe in ['ape', 'learned_ape']:
                pass # APE 和 Learned APE 在输入时直接加到 x 上，不需要单独模块
```

```

# 因果掩码
tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
↪float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
self.register_buffer('mask', mask[None, None, :, :]) # [1, 1, seq_len,
↪seq_len]

def forward(self, x): # x: [batch_size, seq_len, d_model]
    # 线性投影与分头
    batch_size, seq_len, _ = x.shape
    Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # [batch_size, num_heads, seq_len, d_k]
    K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后两维)
    V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 在 seq_len 维度上分配注意力
    # 根据选择的 PE 类型对 Q 和 K 进行位置编码
    for pe_module in self.qk_pe_modules:
        Q = pe_module(Q)
        K = pe_module(K)
    if not self.training:
        # 缩放点积注意力与因果掩码 Causal Masking
        ## Query-Key 点积计算注意力得分
        scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5)
↪# [batch_size, num_heads, seq_len, seq_len]
        if self.sc_pe_modules:
            fused_mask = self.sc_pe_modules[0](seq_len) # [num_heads,
↪seq_len, seq_len]
            scores = scores + fused_mask # 将 ALiBi 的偏置加到得分上
            attention = F.softmax(scores, dim=-1) # [batch_size,
↪num_heads, seq_len, seq_len]
        else:
            ## 因果掩码和 softmax 计算注意力权重

```

```

        attention = F.softmax(scores + self.mask[..., :seq_len, :
↪seq_len], dim=-1) # [batch_size, num_heads, seq_len, seq_len]
        self.captured_attention = attention # ** 新增 **: 捕获注意力矩阵, 用
hook 记录下来, 方便后续分析
        ## 乘以 Value 完成加权求和
        out = torch.matmul(attention, V) # [batch_size, num_heads, ↪
↪seq_len, d_k]
    else:
        if self.sc_pe_modules:
            fused_mask = self.sc_pe_modules[0](seq_len) # [num_heads, ↪
↪seq_len, seq_len]
            out = F.scaled_dot_product_attention(Q, K, V, ↪
↪attn_mask=fused_mask, is_causal=False) # [batch_size, num_heads, seq_len, ↪
↪d_k]
        else:
            out = F.scaled_dot_product_attention(Q, K, V, is_causal=True) ↪
↪# [batch_size, num_heads, seq_len, d_k]
            # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view, 否则会报
            错)
            out = out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1) # ↪
↪[batch_size, seq_len, d_model]
            H = self.W_O(out) # [batch_size, seq_len, d_model]
        return H

```

1.2.3 SwiGLU 和单个 Transformer Block 架构不变

```

[119]: class SwiGLU(nn.Module):
    def __init__(self, d_model, d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model, d_hidden, bias=False)
        self.V = nn.Linear(d_model, d_hidden, bias=False)
        self.W2 = nn.Linear(d_hidden, d_model, bias=False)
    def forward(self, x):
        x1 = F.silu(self.W(x))
        x2 = self.V(x)
        x_out = self.W2(x1 * x2)

```



```
return x_out
```

```
[120]: class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
        b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model) # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model) # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model) # 第一次归一化
            self.norm2 = nn.LayerNorm(d_model) # 第二次归一化
        self.attention = MultiHeadAttention(num_heads, d_model,
            max_seq_len=max_seq_len, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
            head_ratio_upe=head_ratio_upe)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model, d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接
        x1 = x + h
        # 第二次归一化
        x_norm2 = self.norm2(x1)
        # 前馈网络 Feed-Forward Network (FFN)
        x_out = self.ffn(x_norm2)
```

```

# 第二次残差连接
x2 = x1 + x_out
return x2

```

1.2.4 保留 Attention Probe

```

[121]: class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            attention = module.captured_attention.detach().cpu().numpy()
            self.captured_data.append(attention)
        else:
            self.captured_data.append(None) # 如果没有捕获到注意力矩阵, 记录一个
None 占位符
    def reset(self):
        self.captured_data = []

```

1.2.5 Toy Model: Looped Transformer

```

[122]: class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
        ↪b_rope_or_upe=10000, head_ratio_upe=2,
        loop=True, residual_gate=(1,1), residual_gate_type='fixed',
        ↪residual_random=(1,0.1)):
        super().__init__()
        self.loop = loop
        if loop:
            self.transformer_block = TransformerBlock(num_heads, d_model,
            ↪max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type,
            ↪pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
            ↪# 这里保证了各层权重始终相同
            self.probe = AttentionProbe()
            self.transformer_block.attention.register_forward_hook(self.probe)
            self.captured_attention = None

```

```

else:
    self.transformer_block = nn.ModuleList([
        TransformerBlock(num_heads, d_model, max_seq_len=max_seq_len,
        ↪norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type,
        ↪b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
        for _ in range(num_blocks)
    ])
    self.probe = [AttentionProbe() for _ in range(num_blocks)]
    for block, probe in zip(self.transformer_block, self.probe):
        block.attention.register_forward_hook(probe)

    residual_random = torch.tensor(residual_random, dtype=torch.float32)
    if residual_gate == 'random':
        residual_gate_tensor = torch.
        ↪randn(2)*residual_random[1]+residual_random[0] # 随机初始化门控参数，初始正态分
        布
    else:
        residual_gate_tensor = torch.tensor(residual_gate, dtype=torch.
        ↪float32) # 将输入的元组转换为张量，方便后续计算
        if residual_gate_type == 'fixed':
            self.register_buffer('residual_gate', residual_gate_tensor) # 固定
            的门控参数，注册为 buffer 使其成为模型状态的一部分，但不参与梯度更新
        elif residual_gate_type == 'learnable_scalar':
            self.residual_gate = nn.Parameter(torch.
            ↪ones(2)*residual_gate_tensor) # 可学习的标量门控参数，初始值为 1.0
        elif residual_gate_type == 'learnable_vector':
            if residual_gate == 'random':
                self.residual_gate = nn.Parameter(torch.randn(d_model,
                ↪2)*residual_random[1]+residual_random[0]) # 可学习的向量门控参数，初始值为随机
                向量
            else:
                self.residual_gate = nn.Parameter(torch.ones(d_model,
                ↪2)*residual_gate_tensor) # 可学习的向量门控参数，初始值为全 1 向量

    pe_type = self.transformer_block.attention.pe_type
    self.ape = nn.ModuleList()

```

```

for pe in pe_type:
    if pe == 'ape':
        self.ape.append(APE(d_model, max_seq_len=max_seq_len))
    elif pe == 'learned_ape':
        self.ape.append(LearnedAPE(d_model, max_seq_len=max_seq_len))
    elif pe in ['rope', 'ms_upe']:
        pass # RoPE 和 MS_UPE 在 MultiHeadAttention 内部处理位置编码，不需要单独的模块在这里处理

```

```

self.num_blocks = num_blocks
self.num_heads = num_heads
self.d_model = d_model
self.max_seq_len = max_seq_len
self.pe_type = pe_type

```

```

def forward(self, x_0, num_eff=15, current_blocks=None):
    # num_eff 是需要加入误差计算的有效层数，也即理论公式中的  $T$ 
    # x 的形状为 [batch_size, seq_len, d_model]
    a=self.residual_gate[...,0]
    b=self.residual_gate[...,1]
    for ape_module in self.ape:
        x_0 = ape_module(x_0) # 绝对位置编码只加在最开始输入处
    x = x_0
    if current_blocks is None:
        current_blocks = self.num_blocks
    b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对应的起始层索引
    outputs = []
    if self.loop:
        self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向传播的注意力矩阵
        for i in range(current_blocks):
            if i == b_0:
                x= x.detach() # 释放前面的计算图
                x.requires_grad_(True)
                x = self.transformer_block(a*x+b*x_0)

```

```

        if i >= b_0:
            outputs.append(x)
            self.captured_attention = list(self.probe.captured_data) # 显式 copy
一份数据，避免后续被修改
        else:
            for probe in self.probe:
                probe.reset()
            for i, block in enumerate(self.transformer_block):
                if i == b_0:
                    x = x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                    x = block(a*x+b*x_0) # x: [batch_size, seq_len, d_model]
                    if i >= b_0:
                        outputs.append(x)
                        self.captured_attention = [probe.captured_data for probe in self.
→probe]
                outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len,
→d_model]
            return outputs

```

1.2.6 Downstream Task: Linear Regression

转换头

```

[123]: class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1, bias=False, init_scale=None):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model, bias=bias)
        self.read_in_y = nn.Linear(d_y, d_model, bias=bias)
        if init_scale is not None:
            nn.init.normal_(self.read_in_x.weight, mean=0.0, std=init_scale)
            nn.init.normal_(self.read_in_y.weight, mean=0.0, std=init_scale)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半，因为 x 和 y 交织在一起，所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data) # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data) # [batch_size, k, d_model]

```

```

# 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)
stacked = torch.stack((x_emb, y_emb), dim=2) # [batch_size, k, 2,
↪ d_model]

Prompt = stacked.flatten(1,2) # [batch_size, 2k, d_model]

return Prompt

```

损失函数

```

[124]: class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse', weight_decay=1.0):
        super().__init__()
        self.read_out = nn.Linear(d_model, d_y)
        self.weight_decay = weight_decay
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss()
        elif loss_type == 'l1':
            self.loss_fn = nn.L1Loss()

    def forward(self, outputs, y_true, is_eval=False, sink_padding=None):
        # outputs: [batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出
            作为预测值
            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final

            y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2,
↪ d_model]
            y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2,
↪ d_y]

            if sink_padding is not None:
                y_preds = y_preds[:, :, sink_padding:, :]
                y_true = y_true[:, sink_padding:, :]

            loss = self.loss_fn(y_preds, y_true.unsqueeze(1).expand_as(y_preds)) #
↪ [batch_size, num_eff, seq_len/2, d_y]

            return loss

```

拼装

```
[125]: class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
                  norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
    ↪ b_rope_or_upe=10000, head_ratio_upe=2,
                  loop=True, loss_type='mse',
                  residual_gate=(1,1), residual_gate_type='fixed',
    ↪ residual_random=(1,0.1),
                  bias=False, init_scale=None):
        super().__init__()
        self.toy_model = ToyModel(num_blocks, num_heads, d_model, max_seq_len,
                                   norm_type=norm_type, ffn_type=ffn_type,
    ↪ pe_type=pe_type,
                                   loop=loop, b_rope_or_upe=b_rope_or_upe,
    ↪ head_ratio_upe=head_ratio_upe,
                                   residual_gate=residual_gate,
    ↪ residual_gate_type=residual_gate_type, residual_random=residual_random)
        self.head = RegressionHead(d_model, d_x, d_y, bias=bias,
    ↪ init_scale=init_scale)
        self.loss_fn = PredictionLoss(d_model, d_y, loss_type=loss_type)

    def forward(self, x_data, y_data, num_eff, current_blocks=None,
    ↪ is_eval=False, sink_padding=None):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff,
    ↪ current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        output = self.loss_fn(outputs, y_data, is_eval=is_eval,
    ↪ sink_padding=sink_padding)
        return output
```

1.2.7 数据生成与加载

数据生成

线性回归数据

```
[126]: def linear_data_generator(batch_size, seq_len, d_x, d_y, device='cpu'):
    w = torch.randn(batch_size, d_x, d_y, device=device) # 真实线性函数 w:
    ↪ [batch_size, d_x, d_y]
```

```

    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device) # 输入特征
    x: [batch_size, seq_len//2, d_x]
    y_data = x_data @ w # 计算标签 y = x @ w: [batch_size, seq_len//2, d_y]
    return x_data, y_data

```

数据加载器

```

[127]: def dataloader(batch_size=64, seq_len=80, d_x=20, d_y=1, device='cpu',
    ↪data_type='linear', sink_padding=1):
    while True:
        if data_type == 'linear':
            x_data, y_data = linear_data_generator(batch_size, seq_len, d_x,
            ↪d_y, device=device)
            if sink_padding is not None:
                x_data = F.pad(x_data, (0, 0, sink_padding, 0), value=0) # 在
seq_len 维度最前面添加 sink_padding 个全 0 向量
                y_data = F.pad(y_data, (0, 0, sink_padding, 0), value=0)
            yield x_data, y_data

```

1.2.8 实验模块

```

[128]: class LoopedTransformerExperiment:
    def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4,
    ↪gate_lr_ratio=100, max_seq_len=4096, d_x=20, d_y=1,
        seed=None, experiment_name='Experiment',
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
    ↪b_rope_or_upe=10000, head_ratio_upe=2,
        optimizer_type='adam',
        loop=True, loss_type='mse', bias=False, init_scale=None,
        residual_gate=(1,1), residual_gate_type='fixed',
    ↪residual_random=(1,0.1),
        task='regression', timing=True):
        if timing:
            start_time = time.time()
            # 设备选择: 优先使用 MPS (适用于 Apple Silicon), 其次是 CUDA (适用于
    ↪NVIDIA GPU), 最后退回 CPU
            if torch.backends.mps.is_available():
                self.device = torch.device("mps")

```



```

elif torch.cuda.is_available():
    self.device = torch.device("cuda")
else:
    self.device = torch.device("cpu")
self.is_offloaded = False
if seed is not None:
    torch.manual_seed(seed)
# 模型和优化器
self.task = task
if task == 'regression':
    self.model = RegressionSolver(num_blocks, num_heads, d_model,
    ↪d_x=d_x, d_y=d_y, max_seq_len=max_seq_len,
                                norm_type=norm_type,
    ↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
                                head_ratio_upe=head_ratio_upe,
    ↪loop=loop, loss_type=loss_type,
                                bias=bias, init_scale=init_scale,
                                residual_gate=residual_gate,
    ↪residual_gate_type=residual_gate_type, residual_random=residual_random
                                ).to(self.device)

    gate_params=[]
    base_params=[]
    for name, param in self.model.named_parameters():
        if 'residual_gate' in name:
            gate_params.append(param)
        else:
            base_params.append(param)
    param_groups = [{'params': base_params},
                    {'params': gate_params, 'lr': lr * gate_lr_ratio}]
    if optimizer_type == 'adam':
        self.optimizer = torch.optim.Adam(param_groups, lr=lr)
    elif optimizer_type == 'sgd':
        self.optimizer = torch.optim.SGD(param_groups, lr=lr)
    elif optimizer_type == 'adamw':
        self.optimizer = torch.optim.AdamW(param_groups, lr=lr)
self.d_x = d_x

```

```

self.d_y = d_y
self.num_blocks = num_blocks
self.init_time = None
if experiment_name is not None:
    if timing:
        self.init_time = time.time() - start_time
        print(f"{experiment_name} initialized in {self.init_time:.2f}␣
↪seconds. Using device: {self.device}")
    else:
        print(f"{experiment_name} initialized. Using device: {self.
↪device}")

def train(self, batch_size=64, seq_len=80, epochs=20, data_type='linear',
        num_eff=15, sink_padding=None, scheduled_training=True,␣
↪scheduler_type=None,
        print_every=None, timing=False):
    # x_data: [batch_size, k, d_x]
    # y_data: [batch_size, k, d_y]
    self.model.train()
    self.train_time = None
    self.data_loader = dataloader(batch_size, seq_len, self.d_x, self.d_y,␣
↪device=self.device, data_type=data_type, sink_padding=sink_padding)
    self.loss_history = []
    self.residual_gate_history = []
    if scheduler_type == 'cosine':
        scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(self.
↪optimizer, T_max=epochs)
    num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数
    if timing:
        start_time = time.time()
    for epoch in range(epochs):
        self.optimizer.zero_grad()
        x_data, y_data = next(self.data_loader) # 从数据加载器中获取一个批次
的数据
        if scheduled_training:

```

```

        current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训练的
        进行, 逐渐增加参与误差计算的有效层数

    else:
        current_blocks = self.num_blocks
        loss = self.model(x_data, y_data, num_eff=num_eff,
        ↪current_blocks=current_blocks, is_eval=False, sink_padding=sink_padding) # 前
        向传播计算损失 [batch_size, num_eff, seq_len/2, d_y]
        loss = loss.mean() # 对所有维度求平均得到一个标量损失值
        loss.backward() # 反向传播计算梯度
        self.optimizer.step() # 更新模型参数
        if scheduler_type is not None:
            scheduler.step() # 更新学习率
        self.loss_history.append(loss.item()) # 记录损失值
        self.residual_gate_history.append(self.model.toy_model.
        ↪residual_gate.detach().cpu().numpy() if hasattr(self.model.toy_model,
        ↪'residual_gate') else None) # 记录门控参数值
        if print_every is not None and (epoch + 1) % print_every == 0:
            print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f},
            ↪current_blocks: {current_blocks}")
        if timing:
            self.train_time = time.time()-start_time
            print(f"Training completed in {self.train_time:.2f} seconds.")
            self.residual_gate_values = self.residual_gate_history[-1]
            self.final_loss = self.loss_history[-1]

    def evaluate(self, batch_size=64, seq_len=80, data_type='linear',
    ↪loss_type='mse', sink_padding=1):
        self.model.eval()
        with torch.no_grad():
            # 在评估模式下, 直接使用所有层的输出进行预测 (因为已经 no_grad 了, 所以
            不需要防止梯度爆炸了)
            x_eval, y_eval = next(dataloader(batch_size, seq_len+1, self.d_x,
            ↪self.d_y, device=self.device, data_type=data_type,
            ↪sink_padding=sink_padding))

```

```

        y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks,
↪current_blocks=self.num_blocks, is_eval=True, sink_padding=sink_padding) #
↪[batch_size, d_y]

        if loss_type == 'mse':
            mse=F.mse_loss(y_pred, y_eval[:, -1, :])
            print(f"MSELoss: {mse:.4f}") # 计算预测值与真实值之间的均方误差
            return mse.item()

        elif loss_type == 'l1':
            l1=F.l1_loss(y_pred, y_eval[:, -1, :])
            print(f"L1Loss: {l1:.4f}")
            return l1.item()

    def get_results(self, result_list:list[str]):
        all_results = {
            'init_time': self.init_time, #
↪float: 实验初始化时间 (秒)
            'train_time': self.train_time, #
↪float: 训练时间 (秒)
            'loss_history': self.loss_history, #
↪list(float): 训练过程中每个 epoch 的损失值
            'final_loss': self.final_loss, #
↪float: 最终的损失值
            'residual_gate_history': self.residual_gate_history, #
↪list[np.array[2] 或 np.array[d_model, 2]]: 训练过程中每个 epoch 的残差门控参数
值
            'final_residual_gate': self.residual_gate_values, # np.
↪array[2] 或 np.array[d_model, 2]: 最终的残差门控参数值
            'captured_attention': self.model.toy_model.captured_attention #
↪list[np.array[batch_size, num_heads, seq_len, seq_len]]: 捕获的注意力权重
        }

        if any("residual_gate" in key for key in result_list) and self.
↪residual_gate_values is not None:
            if len(self.residual_gate_values.shape) == 1:
                all_results['residual_gate_history_a'] = [gate[0] for gate in
↪self.residual_gate_history]

```

```

        all_results['residual_gate_history_b'] = [gate[1] for gate in
↪self.residual_gate_history]
        all_results['residual_gate_history_a_relative'] = [gate[0]-self.
↪residual_gate_history[0][0] for gate in self.residual_gate_history]
        all_results['residual_gate_history_b_relative'] = [gate[1]-self.
↪residual_gate_history[0][1] for gate in self.residual_gate_history]
        all_results['final_residual_gate_a'] = self.
↪residual_gate_values[0]
        all_results['final_residual_gate_b'] = self.
↪residual_gate_values[1]
        elif len(self.residual_gate_values.shape) == 2:
            all_results['residual_gate_history_a_mean'] = [gate[:,0].mean()
↪for gate in self.residual_gate_history]
            all_results['residual_gate_history_b_mean'] = [gate[:,1].mean()
↪for gate in self.residual_gate_history]
            all_results['residual_gate_history_a_mean_relative'] = [gate[:
↪,0].mean()-self.residual_gate_history[0][:,0].mean() for gate in self.
↪residual_gate_history]
            all_results['residual_gate_history_b_mean_relative'] = [gate[:
↪,1].mean()-self.residual_gate_history[0][:,1].mean() for gate in self.
↪residual_gate_history]
            all_results['residual_gate_history_a_std'] = [gate[:,0].std()
↪for gate in self.residual_gate_history]
            all_results['residual_gate_history_b_std'] = [gate[:,1].std()
↪for gate in self.residual_gate_history]
            all_results['final_residual_gate_a_mean'] = self.
↪residual_gate_values[:,0].mean()
            all_results['final_residual_gate_b_mean'] = self.
↪residual_gate_values[:,1].mean()
            all_results['final_residual_gate_a_std'] = self.
↪residual_gate_values[:,0].std()
            all_results['final_residual_gate_b_std'] = self.
↪residual_gate_values[:,1].std()
            invalid_keys = [key for key in result_list if key not in all_results]
            if invalid_keys:

```

```

        print(f"Warning: The following requested results are not available_
↪and will be ignored: {invalid_keys}")

        if result_list is not None:
            return {key: all_results[key] for key in result_list if key in_
↪all_results}

def offload_to_cpu(self):
    self.model.to('cpu')
    for state in self.optimizer.state.values():
        for k, v in state.items():
            if isinstance(v, torch.Tensor):
                state[k] = v.cpu()
    self.is_offloaded = True

def load_to_device(self):
    self.model.to(self.device)
    for state in self.optimizer.state.values():
        for k, v in state.items():
            if isinstance(v, torch.Tensor):
                state[k] = v.to(self.device)
    self.is_offloaded = False

```

1.2.9 默认配置参数

```

[ ]: def default_setup(manual=None):
    init_parameters = {
        # RoPE/MS-UPE
        'b_rope_or_upe': 10000,          # float: RoPE 或 UPE 的基数
        'head_ratio_upe': 2,             # float: UPE 头比例

        # MultiHeadAttention
        'num_heads': 8,                  # int: 注意力头数 H
        'd_model': 256,                  # int: Transformer 的维度 D
        'max_seq_len': 100,              # int: 模型支持的最大序列长度（位置编码
        # 相关）
        'pe_type': ['learned_ape'],      # list: 位置编码类型 ('ape',_
        ↪'learned_ape', 'rope', 'ms_upe', 'alibi')
    }

```

```

# TransformerBlock
    'norm_type': 'layernorm',          # str: transformer_block 中的归一化类
型 ('layernorm' 或 'rmsnorm')
    'ffn_type': 'gelu',                # str: transformer_block 中前馈网络激
活函数类型 ('gelu' 或 'swiglu')

# ToyModel
    'num_blocks': 20,                  # int: Transformer 的层数 b
    'loop': True,                      # bool: 是否权重共享
    'residual_gate': (1,1),            # tuple or str: transformer_block 之
间传递残差的门控参数初始值, 使  $x=a*x+b*x_0$  ((a,b) 或 'random')
    'residual_gate_type': 'fixed',     # str: 残差门控类型 ('fixed',
↪ 'learnable_scalar', 'learnable_vector')
    'residual_random': (1,0.1),        # tuple (mean, std): 当
↪ residual_gate='random' 时满足高斯分布  $N(\text{mean}, \text{std})$ 

# RegressionHead
    'd_x': 20,                         # int: 数据输入的特征维度
    'd_y': 1,                         # int: 数据输出的特征维度
    'bias': False,                    # bool: Regression Head 是否使用偏置
    'init_scale': None,               # float or None: Regression Head 的初
始化缩放因子 (None 表示使用默认初始化)

# PredictionLoss
    'loss_type': 'mse',                # str: 损失函数类型 ('mse' 或 'l1')

# LoopedTransformerExperiment
    'lr': 1e-4,                       # float: 学习率
    'gate_lr_ratio': 100,              # float: 当使用残差门控时, 门控参数的
学习率相对于模型其他参数的倍数
    'seed': 42,                       # int or None: 手动随机种子, 确保实验
可复现
    'experiment_name': 'Experiment',   # str or None: 实验名称 (None 表示不打
印设备和实验初始化信息)

```

```

        'timing': True,                                # bool: 是否打印初始化时间
        'optimizer_type': 'adam',                     # str: 优化器类型 ('adam' 或 'sgd'或
↪ 'adamw')
        'task': 'regression',                         # str: 任务类型 ('regression')
    }

    train_parameters = {
        # MultiHeadAttention
        'batch_size': 64,                             # int: 数据批次大小
        'seq_len': 80,                                # int: 数据序列长度

        # ToyModel
        'num_eff': 15,                                # int: 有效层数, 即参与误差计算、参与梯
度反向传播的层数  $T=b-b_0$ 

        # dataloader
        'sink_padding': None,                         # int or None: seq 最前面填充的 sink_
↪ token 组数 (None 表示不使用 sink padding)

        # LoopedTransformerExperiment
        'epochs': 20,                                 # int: 训练轮数
        'data_type': 'linear',                        # str: 回归数据类型 ('linear')
        'scheduler_type': None,                      # str or None: 学习率调节器类型
↪ ('cosine' 或 None)
        'scheduled_training': True,                  # bool: 是否使用计划训练, 即随着训练的
进行逐渐增加参与的层数  $b=epoch+num\_eff$  , 直到达到最大层数  $num\_blocks$ 
        'print_every': 5,                            # int or None: 打印频率/epoch (None
表示不打印训练过程中的损失信息)
        'timing': True,                              # bool: 是否打印训练时间
    }

    if manual is not None:
        for key,value in manual.items():
            if key in init_parameters:
                init_parameters[key]=value
            elif key in train_parameters:
                train_parameters[key]=value

```



```

        else:
            print(f"Warning:key {key} do not exist in init_parameters and_
↪train_parameters.")
    return init_parameters, train_parameters

```

1.2.10 显存检测

```

[130]: def print_vram_usage(tag=""):
    if torch.backends.mps.is_available():
        # MPS 专用 API (Apple Silicon)
        allocated = torch.mps.current_allocated_memory() / (1024 ** 3)
        driver_alloc = torch.mps.driver_allocated_memory() / (1024 ** 3)
        print(f"[{tag}] 显存占用 -> PyTorch 分配: {allocated:.2f} GB | Metal 驱动
↪分配: {driver_alloc:.2f} GB")

    elif torch.cuda.is_available():
        # CUDA 专用 API
        allocated = torch.cuda.memory_allocated() / (1024 ** 3)
        reserved = torch.cuda.memory_reserved() / (1024 ** 3)
        total = torch.cuda.get_device_properties(0).total_memory / (1024 ** 3)
        free = total - reserved
        print(f"[{tag}] 显存占用 -> 已分配: {allocated:.2f} GB | 缓存池:
↪{reserved:.2f} GB | 剩余可用: {free:.2f} GB")

    else:
        print(f"[{tag}] 当前运行在 CPU, 占用主板内存。")

```

1.3 实验台搭建

1.3.1 传入参数、获得结果、画图

```

[ ]: class ExperimentTable:
    '''
    Attributes:
        num_experiments: int, 实验数量
        init_parameters: list of dict, 在 __init__() 方法中传入, 用于指定每个实验
        的初始化参数, 可以覆盖默认参数
    '''

```

train_parameters: list of dict, 在 `__init__()` 方法中传入, 用于指定每个实验的训练参数, 可以覆盖默认参数

result_lists: list of tuple of list, 在 `run()` 方法中传入, 用于指定需要展示的结果名称和对应的横坐标类型。

results_groups: list of tuple of list, 包含不同模式 (如 'train' 和 'evaluate') 下的结果列表, 每个结果列表是一个包含所有实验结果的列表,

如 `[('train', [experiment1_train_results, experiment2_train_results, ...]), ('evaluate', [experiment1_evaluate_results, experiment2_evaluate_results, ...])]`

'''

```
def __init__(self, params_groups:list[dict], manual=None):
```

'''

Args:

params_groups: list of dict, 每个 *dict* 包含一个实验的参数设置, 可以覆盖默认参数,

如 `[{residual_gate: (0.5, 0.5), experiment_name: 'Experiment 1'}, {residual_gate: (0.8, 0.2), experiment_name: 'Experiment 2'}]`

manual: dict, 全局默认参数修改

Attributes:

num_experiments: int, 实验数量

init_parameters: list of dict, 每个 *dict* 包含一个实验的初始化参数设置

train_parameters: list of dict, 每个 *dict* 包含一个实验的训练参数设置

'''

```
init, train = default_setup(manual=manual)
```

```
self.num_experiments = len(params_groups)
```

```
self.init_parameters = [copy.deepcopy(init) for _ in range(self.  
num_experiments)]
```

```
self.train_parameters = [copy.deepcopy(train) for _ in range(self.  
num_experiments)]
```

```
# 自动分类传入的参数以更新默认参数
```

```
for i, params in enumerate(params_groups):
```

```
    for key, value in params.items():
```

```
        if key in self.init_parameters[i]:
```

```

        self.init_parameters[i][key] = value
    elif key in self.train_parameters[i]:
        self.train_parameters[i][key] = value
    else:
        raise ValueError(f"Unknown parameter: {key}")

    def run(self, result_lists: list[tuple[list[str], str]], modes=['train'],
    ↪parallel_workers=1):
        '''
        Args:
            result_lists: list of tuple of list, 每个 tuple 包含一组实验测量的数
            据和对应的横坐标类型, 每个 list 中的字符串表示需要展示的结果名称, 必须是
            ↪LoopedTransformerExperiment.get_results() 中返回的结果名称,
            如 [(['loss_history', 'residual_gate_history_a_mean'],
            ↪'residual_gate_history_b_mean'], 'epoch'), (['final_loss',
            ↪'final_residual_gate_a_mean', 'final_residual_gate_b_mean'], 'experiment')]

            modes: list of str, 表示需要运行的模式, 可以是 'train' 和 'evaluate'
            ↪的任意组合, 'train' 表示需要训练模型并记录训练结果, 'evaluate' 表示需要在训练完成
            后进行评估并记录评估结果

            parallel_workers: int, 表示并行运行实验的工作线程数。

        Attributes:
            result_lists: 同 Args
            results_groups: dict, 包含不同模式 (如 'train' 和 'evaluate') 下的结果列
            表, 每个 value 是一个 list of dict, 每个 dict 是一个实验的结果, 为 metric 和 value
            的键值对。

            experiments: list of LoopedTransformerExperiment 对象, 已转移到 CPU
            以节省 GPU 显存
        '''
        # 创建实验对象并运行
        result_set = set().union(*[result_list for result_list, _ in
    ↪result_lists])
        self.result_lists = result_lists

```

```

        self.results_groups = {'train': [None]*self.num_experiments, 'evaluate':
↪ [None]*self.num_experiments}

        self.experiments=[None]*self.num_experiments
        print_lock = threading.Lock() # 用于在线程中安全地打印日志
        vram_printed = False
        def run_single_experiment(i, parallel=False):
            nonlocal vram_printed
            if parallel:
                self.init_parameters[i]['timing']=False
                self.train_parameters[i]['timing']=False
                self.train_parameters[i]['print_every']=None
            experiment = LoopedTransformerExperiment(**self.init_parameters[i])
            if parallel:
                with print_lock:
                    if not vram_printed:
                        print_vram_usage(tag=f"Parallel 峰值显存检测")
                        vram_printed = True
            else:
                print_vram_usage(tag=f"Experiment {i+1} 训练前")
            for mode in modes:
                if mode == 'train':
                    experiment.train(**self.train_parameters[i])
                    self.results_groups['train'][i] = experiment.
↪get_results(result_set)
                    elif mode == 'evaluate':
                        eval_parameters = {k:v for k,v in self.train_parameters[i].
↪items() if k in ['batch_size', 'seq_len', 'data_type', 'sink_padding']}
                        if 'loss_type' in self.init_parameters[i]:
                            eval_parameters['loss_type'] = self.
↪init_parameters[i]['loss_type'] # 确保评估使用与训练相同类型的损失函数
                        self.results_groups['evaluate'][i] = experiment.
↪evaluate(**eval_parameters)
                        experiment.offload_to_cpu() # 训练完成后将模型和结果移回 CPU, 节省 GPU
显存

                self.experiments[i] = experiment
                if torch.cuda.is_available():

```

```

        torch.cuda.empty_cache()
    if torch.backends.mps.is_available():
        torch.mps.empty_cache()
    if not parallel:
        print_vram_usage(tag=f"Experiment {i+1} 清理后")
        print(f"Experiment {i+1}/{self.num_experiments} completed.\n")

print_vram_usage(tag=f"本底显存检测")
if parallel_workers == 1:
    for i in range(self.num_experiments):
        run_single_experiment(i, parallel=False)
    elif isinstance(parallel_workers, int) and parallel_workers > 1:
        print(f"Running {self.num_experiments} experiments in parallel with
↳{parallel_workers} workers...")
        with concurrent.futures.
↳ThreadPoolExecutor(max_workers=parallel_workers) as executor:
            executor.map(lambda i: run_single_experiment(i, parallel=True),
↳range(self.num_experiments))
    else:
        raise ValueError("parallel_workers must be a positive integer.")

def plot(self, compare_experiments=True,
        subplot_shape=(1,-1),figure_size=(15,10),suptitle='Looped
↳Transformer Experiment Results',
        colors=['blue', 'red', 'green', 'yellow', 'cyan', 'orange',
↳'purple', 'brown']):
    '''

```

Args:

compare_experiments: bool, True 表示在一张图中比较不同实验的结果，每张图仅比较一个指标；*False* 表示每个实验单独成图，每张图仅比较一个实验的多个指标（第一个指标使用左轴，第二个及以后的指标使用共享 x 轴的右轴）

subplot_shape: tuple of int, 表示子图的行数和列数，(-1 表示自动计算)，如 (1, -1) 表示所有图在一行，(-1, 2) 表示每行两个图，自动计算行数

figure_size: tuple of float, 画布大小，单位为英寸，如 (15, 10)

```

        supitle: str, 整体图的标题

        colors: list of str, 画图使用的颜色列表, 每张图从头开始循环使用这些颜色
    """
    # 画图
    if compare_experiments:
        num_plots = len(self.result_lists) # 需要画的图的总数量
    else:
        num_plots = len(self.result_lists)*self.num_experiments # 需要画的图
的总数量

    if subplot_shape == (-1, -1):
        raise ValueError("Invalid subplot_shape: both dimensions cannot be -1. Please specify at least one dimension or provide a valid shape.")
    if subplot_shape[1] == -1:
        subplot_shape = (subplot_shape[0], math.ceil(num_plots / subplot_shape[0])) # 自动计算列数
    elif subplot_shape[0] == -1:
        subplot_shape = (math.ceil(num_plots / subplot_shape[1]), subplot_shape[1]) # 自动计算行数
    if subplot_shape[0] * subplot_shape[1] < num_plots:
        raise ValueError(f"subplot_shape {subplot_shape} is too small for the number of plots {num_plots}. Please increase the dimensions.")

    if self.results_groups['train'][0] is not None:
        results = self.results_groups['train']
        fig, axes = plt.subplots(*subplot_shape, figsize=figure_size)
        axes_flat = np.atleast_1d(axes).flatten() # 将 axes 转换为一维数组, 方便索引
        index=0
        for result_list, x in self.result_lists:
            if x=='epoch': # 表示画横坐标是 epoch 的曲线
                if compare_experiments:
                    for metric in result_list:
                        for i in range(self.num_experiments):

```

```

exp_name = self.init_parameters[i].
get('experiment_name', f'Experiment {i+1}')
    axs_flat[index].plot(results[i][metric],
label=exp_name, color=colors[i%len(colors)])
    axs_flat[index].set_xlabel('Epoch')
    axs_flat[index].set_ylabel(metric)
    axs_flat[index].tick_params(axis='y')
    axs_flat[index].set_title('Comparison of ' + metric)
    axs_flat[index].legend(loc='upper right')
    axs_flat[index].grid(True)
    index += 1
else:
    for i in range(self.num_experiments):
        color_index = 0
        axs_flat[index].plot(results[i][result_list[0]],
label=result_list[0], color=colors[color_index%len(colors)])
        axs_flat[index].set_xlabel('Epoch')
        axs_flat[index].set_ylabel(result_list[0],
color=colors[color_index%len(colors)])
        axs_flat[index].tick_params(axis='y',
labelcolor=colors[color_index%len(colors)])
        color_index += 1
        if len(result_list) > 1:
            ax_twin = axs_flat[index].twinx() # 创建共享 x
轴但独立 y 轴的第二个坐标轴
            primary_color = colors[color_index%len(colors)]
            for j in range(1, len(result_list)):
                ax_twin.plot(results[i][result_list[j]],
label=result_list[j], color=colors[color_index%len(colors)])
                color_index += 1
            ax_twin.set_ylabel(' and '.join(result_list[1:
]), color=primary_color)
            ax_twin.tick_params(axis='y',
labelcolor=primary_color)
            ax_twin.legend(loc='upper right')

```

```

        axs_flat[index].set_title(self.
↪init_parameters[i]['experiment_name']+'-'+ and '.join(result_list))
        axs_flat[index].legend(loc='upper left')
        axs_flat[index].grid(True)
        index += 1

    elif x=='experiment': # 表示画不同实验对比的柱状图
        pass
    # 如果绘制的图的数量少于子图的总数量，则隐藏多余的子图
    for i in range(index, len(axs_flat)):
        axs_flat[i].set_visible(False)
    if self.results_groups['evaluate'][0] is not None:
        pass
    if supitle is not None:
        plt.supitle(supitle)
    plt.tight_layout()
    plt.show()

```

1.4 主程序

1.4.1 实验数据观测

绘制 Loss 下降曲线

```

[137]: experiment_table = ExperimentTable(params_groups=[{}])
        experiment_table.run(result_lists=[(['loss_history'], 'epoch')])
        experiment_table.plot(figsize=(8,6))

```

[本底显存检测] 显存占用 -> PyTorch 分配: 0.12 GB | Metal 驱动分配: 0.16 GB

Experiment initialized in 0.02 seconds. Using device: mps

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.12 GB | Metal 驱动分配: 0.16 GB

Epoch 5/20, Loss: 72.5179, current_blocks: 19

Epoch 10/20, Loss: 35.3423, current_blocks: 20

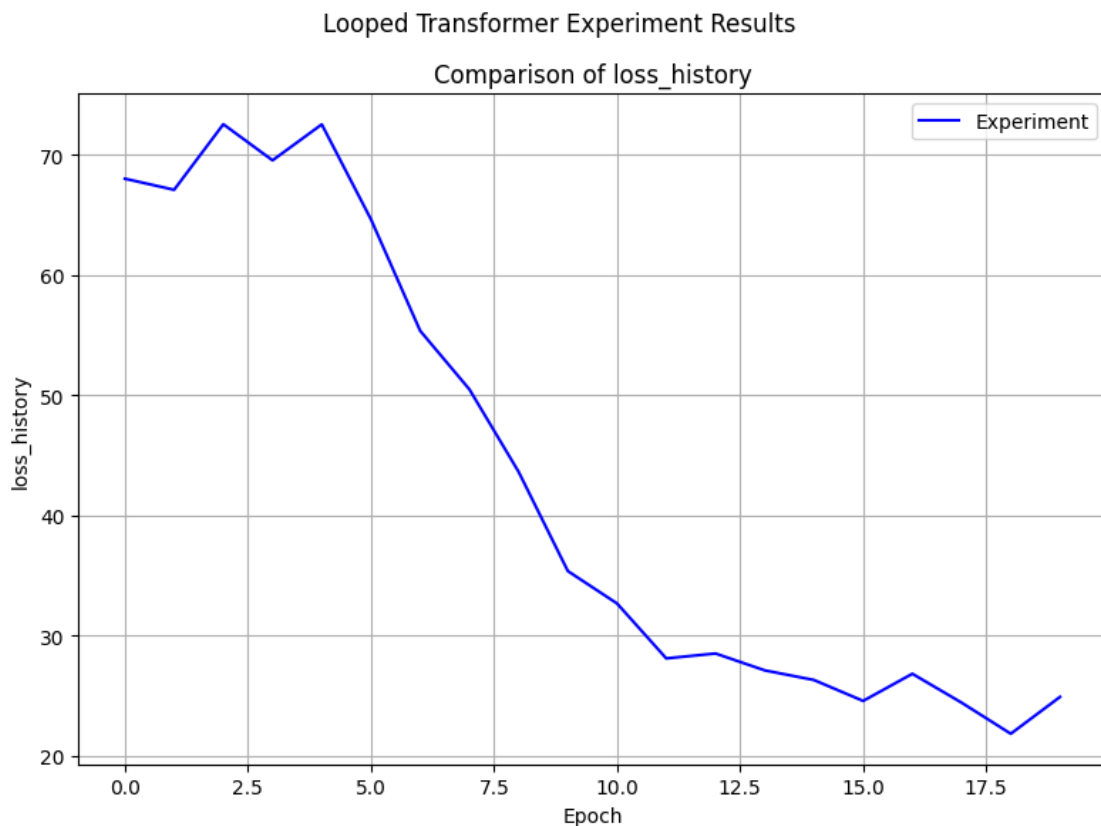
Epoch 15/20, Loss: 26.2775, current_blocks: 20

Epoch 20/20, Loss: 24.8684, current_blocks: 20

Training completed in 6.57 seconds.

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.12 GB | Metal 驱动分配: 0.16 GB

Experiment 1/1 completed.



1.4.2 对比实验

Scheduled Training vs Non-Scheduled Training 对比实验

```
[138]: init_parameters, train_parameters = default_setup()
init_parameters['experiment_name'] = 'Experiment scheduled'
experiment1 = LoopedTransformerExperiment(**init_parameters)
experiment1.train(**train_parameters)

init_parameters['experiment_name'] = 'Experiment non-scheduled'
train_parameters['scheduled_training'] = False
experiment2 = LoopedTransformerExperiment(**init_parameters)
experiment2.train(**train_parameters)

plt.plot(experiment1.loss_history, label='Scheduled Training', color='blue')
plt.plot(experiment2.loss_history, label='Non-Scheduled Training', color='red')
plt.xlabel('Epoch')
```

```
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Scheduled Training vs Non-Scheduled Training')
plt.grid()
plt.legend()
plt.show()
```

Experiment scheduled initialized in 0.02 seconds. Using device: mps

Epoch 5/20, Loss: 72.5179, current_blocks: 19

Epoch 10/20, Loss: 35.3423, current_blocks: 20

Epoch 15/20, Loss: 26.2775, current_blocks: 20

Epoch 20/20, Loss: 24.8684, current_blocks: 20

Training completed in 6.67 seconds.

Experiment non-scheduled initialized in 0.01 seconds. Using device: mps

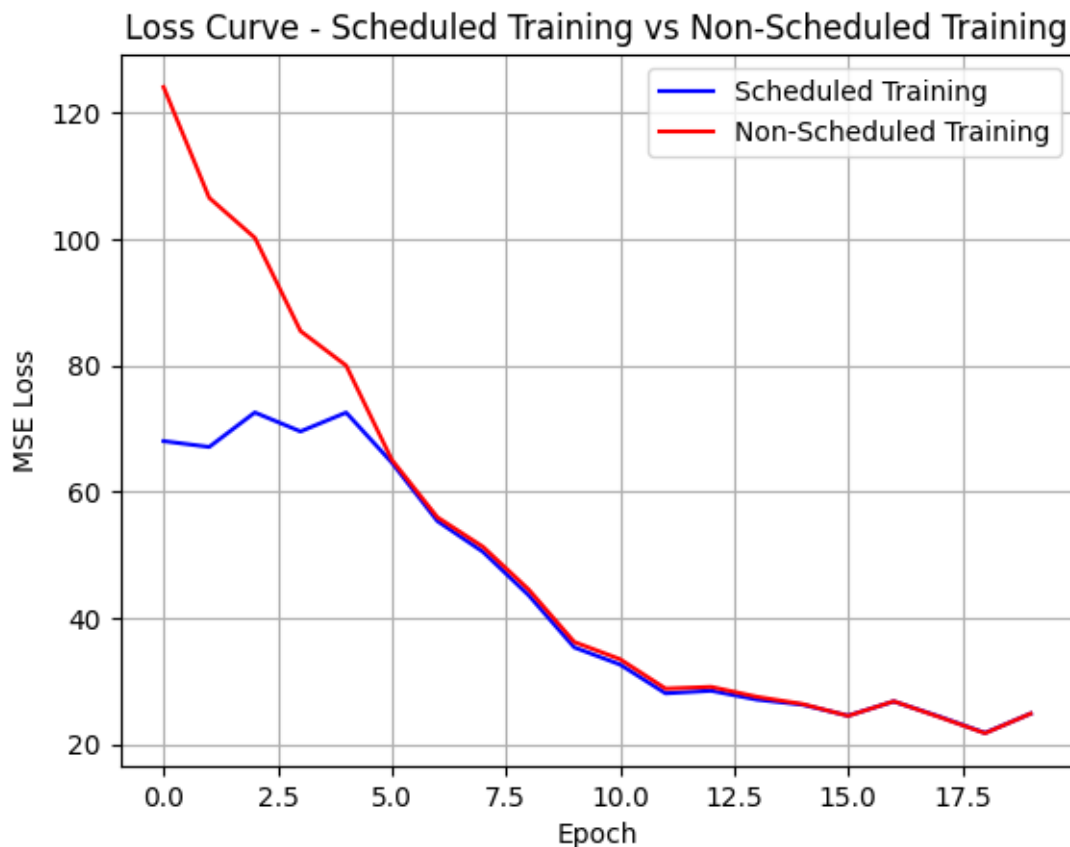
Epoch 5/20, Loss: 79.9593, current_blocks: 20

Epoch 10/20, Loss: 36.1977, current_blocks: 20

Epoch 15/20, Loss: 26.3801, current_blocks: 20

Epoch 20/20, Loss: 24.8631, current_blocks: 20

Training completed in 6.72 seconds.



```
[141]: table = ExperimentTable(params_groups=[{'scheduled_training': True,
↪ 'experiment_name': 'Scheduled Training'},
{'scheduled_training': False,
↪ 'experiment_name': 'Non-Scheduled Training'}])
table.run(result_lists=[(['loss_history'], 'epoch')], parallel_workers=2)
table.plot(figure_size=(8,6), compare_experiments=True)
```

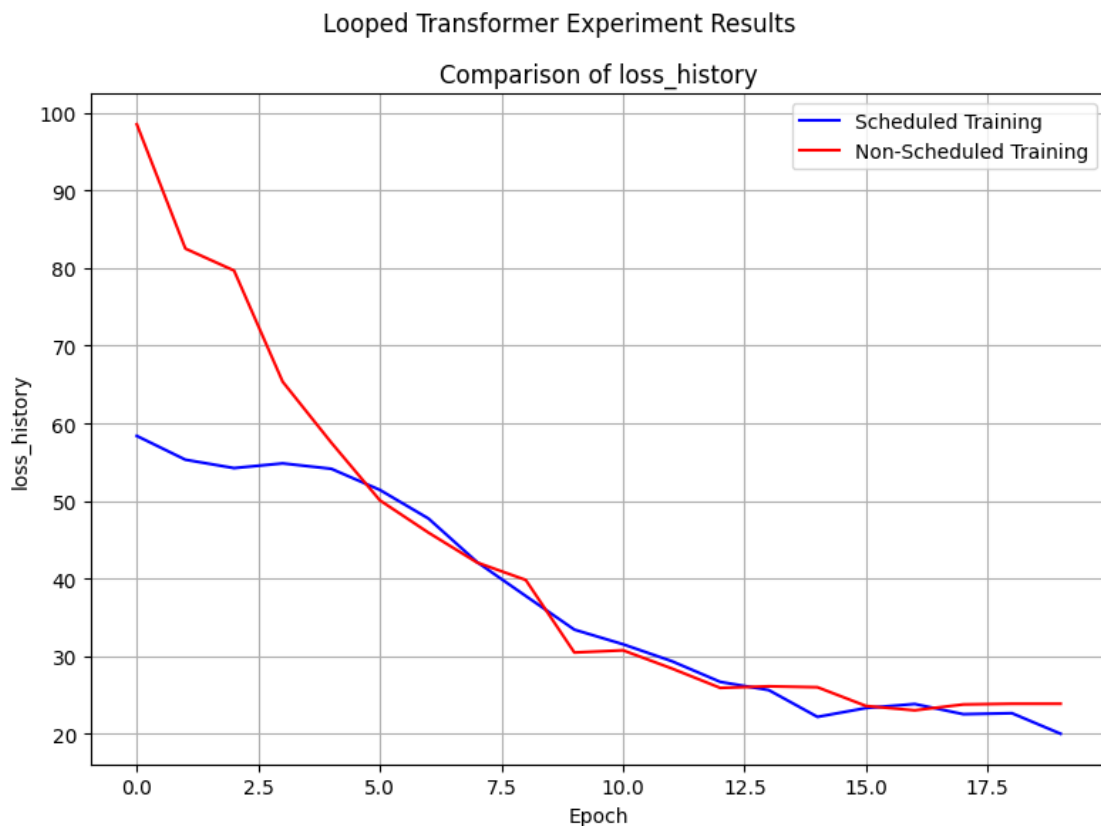
[本底显存检测] 显存占用 -> PyTorch 分配: 0.15 GB | Metal 驱动分配: 0.18 GB

Running 2 experiments in parallel with 2 workers...

Scheduled Training initialized. Using device: mps

[Parallel 峰值显存检测] 显存占用 -> PyTorch 分配: 0.16 GB | Metal 驱动分配: 0.18 GB

Non-Scheduled Training initialized. Using device: mps



不同 Positional Embedding 对比实验

初始参数

```
[ ]: init_parameters, train_parameters = default_setup()
```

各个实验

```
[ ]: init_parameters['experiment_name'] = 'learning_ape_exp'
learning_ape_exp = LoopedTransformerExperiment(**init_parameters)
learning_ape_exp.train(**train_parameters)
```

```
learned_ape_exp initialized. Using device: mps
Epoch 5/100, Loss: 41.1805, current_blocks: 19
Epoch 10/100, Loss: 18.3069, current_blocks: 20
Epoch 15/100, Loss: 7.0673, current_blocks: 20
Epoch 20/100, Loss: 5.1770, current_blocks: 20
Epoch 25/100, Loss: 5.9387, current_blocks: 20
```

```
Epoch 30/100, Loss: 4.4391, current_blocks: 20
Epoch 35/100, Loss: 3.7256, current_blocks: 20
Epoch 40/100, Loss: 3.1097, current_blocks: 20
Epoch 45/100, Loss: 3.2062, current_blocks: 20
Epoch 50/100, Loss: 2.9484, current_blocks: 20
Epoch 55/100, Loss: 2.6715, current_blocks: 20
Epoch 60/100, Loss: 2.9714, current_blocks: 20
Epoch 65/100, Loss: 2.3141, current_blocks: 20
Epoch 70/100, Loss: 2.6434, current_blocks: 20
Epoch 75/100, Loss: 2.3794, current_blocks: 20
Epoch 80/100, Loss: 2.3924, current_blocks: 20
Epoch 85/100, Loss: 2.8761, current_blocks: 20
Epoch 90/100, Loss: 3.1192, current_blocks: 20
Epoch 95/100, Loss: 3.2056, current_blocks: 20
Epoch 100/100, Loss: 2.6971, current_blocks: 20
Training completed in 34.92 seconds.
```

```
[87]: init_parameters['experiment_name'] = 'ape_exp'
      init_parameters['pe_type'] = 'ape'
      ape_exp = LoopedTransformerExperiment(**init_parameters)
      ape_exp.train(**train_parameters)
```

```
ape_exp initialized. Using device: mps
Epoch 5/100, Loss: 10.5527, current_blocks: 19
Epoch 10/100, Loss: 9.5900, current_blocks: 20
Epoch 15/100, Loss: 6.6546, current_blocks: 20
Epoch 20/100, Loss: 5.5995, current_blocks: 20
Epoch 25/100, Loss: 4.4722, current_blocks: 20
Epoch 30/100, Loss: 3.7418, current_blocks: 20
Epoch 35/100, Loss: 3.0496, current_blocks: 20
Epoch 40/100, Loss: 2.7531, current_blocks: 20
Epoch 45/100, Loss: 2.7307, current_blocks: 20
Epoch 50/100, Loss: 2.3372, current_blocks: 20
Epoch 55/100, Loss: 2.0070, current_blocks: 20
Epoch 60/100, Loss: 2.4516, current_blocks: 20
Epoch 65/100, Loss: 1.9479, current_blocks: 20
Epoch 70/100, Loss: 2.1836, current_blocks: 20
```

```
Epoch 75/100, Loss: 1.9636, current_blocks: 20
Epoch 80/100, Loss: 2.0577, current_blocks: 20
Epoch 85/100, Loss: 2.4641, current_blocks: 20
Epoch 90/100, Loss: 2.7406, current_blocks: 20
Epoch 95/100, Loss: 2.6699, current_blocks: 20
Epoch 100/100, Loss: 2.2478, current_blocks: 20
Training completed in 33.32 seconds.
```

```
[88]: init_parameters['experiment_name'] = 'alibi_exp'
      init_parameters['pe_type'] = 'alibi'
      alibi_exp = LoopedTransformerExperiment(**init_parameters)
      alibi_exp.train(**train_parameters)
```

```
alibi_exp initialized. Using device: mps
Epoch 5/100, Loss: 7.9440, current_blocks: 19
Epoch 10/100, Loss: 5.0251, current_blocks: 20
Epoch 15/100, Loss: 4.0158, current_blocks: 20
Epoch 20/100, Loss: 3.2134, current_blocks: 20
Epoch 25/100, Loss: 3.3868, current_blocks: 20
Epoch 30/100, Loss: 2.5129, current_blocks: 20
Epoch 35/100, Loss: 2.3932, current_blocks: 20
Epoch 40/100, Loss: 2.3417, current_blocks: 20
Epoch 45/100, Loss: 2.4873, current_blocks: 20
Epoch 50/100, Loss: 1.9661, current_blocks: 20
Epoch 55/100, Loss: 1.7287, current_blocks: 20
Epoch 60/100, Loss: 2.0730, current_blocks: 20
Epoch 65/100, Loss: 1.7679, current_blocks: 20
Epoch 70/100, Loss: 2.1053, current_blocks: 20
Epoch 75/100, Loss: 1.9362, current_blocks: 20
Epoch 80/100, Loss: 1.7442, current_blocks: 20
Epoch 85/100, Loss: 2.1672, current_blocks: 20
Epoch 90/100, Loss: 2.5389, current_blocks: 20
Epoch 95/100, Loss: 2.2342, current_blocks: 20
Epoch 100/100, Loss: 2.1209, current_blocks: 20
Training completed in 34.19 seconds.
```

```
[89]: init_parameters['experiment_name'] = 'rope_exp'
init_parameters['pe_type'] = 'rope'
init_parameters['b_rope_or_upe'] = 10000
rope_exp = LoopedTransformerExperiment(**init_parameters)
rope_exp.train(**train_parameters)
```

```
rope_exp initialized. Using device: mps
Epoch 5/100, Loss: 8.5121, current_blocks: 19
Epoch 10/100, Loss: 5.3167, current_blocks: 20
Epoch 15/100, Loss: 3.9336, current_blocks: 20
Epoch 20/100, Loss: 3.2812, current_blocks: 20
Epoch 25/100, Loss: 3.6321, current_blocks: 20
Epoch 30/100, Loss: 2.6115, current_blocks: 20
Epoch 35/100, Loss: 2.4616, current_blocks: 20
Epoch 40/100, Loss: 2.4041, current_blocks: 20
Epoch 45/100, Loss: 2.5846, current_blocks: 20
Epoch 50/100, Loss: 2.0036, current_blocks: 20
Epoch 55/100, Loss: 1.7312, current_blocks: 20
Epoch 60/100, Loss: 2.1332, current_blocks: 20
Epoch 65/100, Loss: 1.7742, current_blocks: 20
Epoch 70/100, Loss: 2.0761, current_blocks: 20
Epoch 75/100, Loss: 1.9971, current_blocks: 20
Epoch 80/100, Loss: 1.7743, current_blocks: 20
Epoch 85/100, Loss: 2.2035, current_blocks: 20
Epoch 90/100, Loss: 2.6322, current_blocks: 20
Epoch 95/100, Loss: 2.3430, current_blocks: 20
Epoch 100/100, Loss: 2.2058, current_blocks: 20
Training completed in 46.83 seconds.
```

```
[90]: init_parameters['experiment_name'] = 'ms_upe_exp'
init_parameters['pe_type'] = 'ms_upe'
init_parameters['b_rope_or_upe'] = 10000
init_parameters['head_ratio_upe'] = 2
ms_upe_exp = LoopedTransformerExperiment(**init_parameters)
ms_upe_exp.train(**train_parameters)
```

```
ms_upe_exp initialized. Using device: mps
Epoch 5/100, Loss: 8.0830, current_blocks: 19
```

```

Epoch 10/100, Loss: 5.2086, current_blocks: 20
Epoch 15/100, Loss: 4.1812, current_blocks: 20
Epoch 20/100, Loss: 2.9801, current_blocks: 20
Epoch 25/100, Loss: 3.2729, current_blocks: 20
Epoch 30/100, Loss: 2.6667, current_blocks: 20
Epoch 35/100, Loss: 2.5216, current_blocks: 20
Epoch 40/100, Loss: 2.3067, current_blocks: 20
Epoch 45/100, Loss: 2.4673, current_blocks: 20
Epoch 50/100, Loss: 2.0190, current_blocks: 20
Epoch 55/100, Loss: 1.7739, current_blocks: 20
Epoch 60/100, Loss: 2.1786, current_blocks: 20
Epoch 65/100, Loss: 1.8399, current_blocks: 20
Epoch 70/100, Loss: 2.0640, current_blocks: 20
Epoch 75/100, Loss: 2.0052, current_blocks: 20
Epoch 80/100, Loss: 1.8559, current_blocks: 20
Epoch 85/100, Loss: 2.2586, current_blocks: 20
Epoch 90/100, Loss: 2.6707, current_blocks: 20
Epoch 95/100, Loss: 2.3728, current_blocks: 20
Epoch 100/100, Loss: 2.3211, current_blocks: 20
Training completed in 39.21 seconds.

```

```

[91]: init_parameters['experiment_name'] = 'ms_upe&learned_ape_exp'
      init_parameters['pe_type'] = ['ms_upe', 'learned_ape']
      init_parameters['b_rope_or_upe'] = 10000
      init_parameters['head_ratio_upe'] = 2
      ms_upe_learned_ape_exp = LoopedTransformerExperiment(**init_parameters)
      ms_upe_learned_ape_exp.train(**train_parameters)

```

ms_upe&learned_ape_exp initialized. Using device: mps

```

Epoch 5/100, Loss: 41.0124, current_blocks: 19
Epoch 10/100, Loss: 17.5966, current_blocks: 20
Epoch 15/100, Loss: 6.7853, current_blocks: 20
Epoch 20/100, Loss: 5.4592, current_blocks: 20
Epoch 25/100, Loss: 6.0685, current_blocks: 20
Epoch 30/100, Loss: 4.3091, current_blocks: 20
Epoch 35/100, Loss: 3.6125, current_blocks: 20
Epoch 40/100, Loss: 3.1510, current_blocks: 20

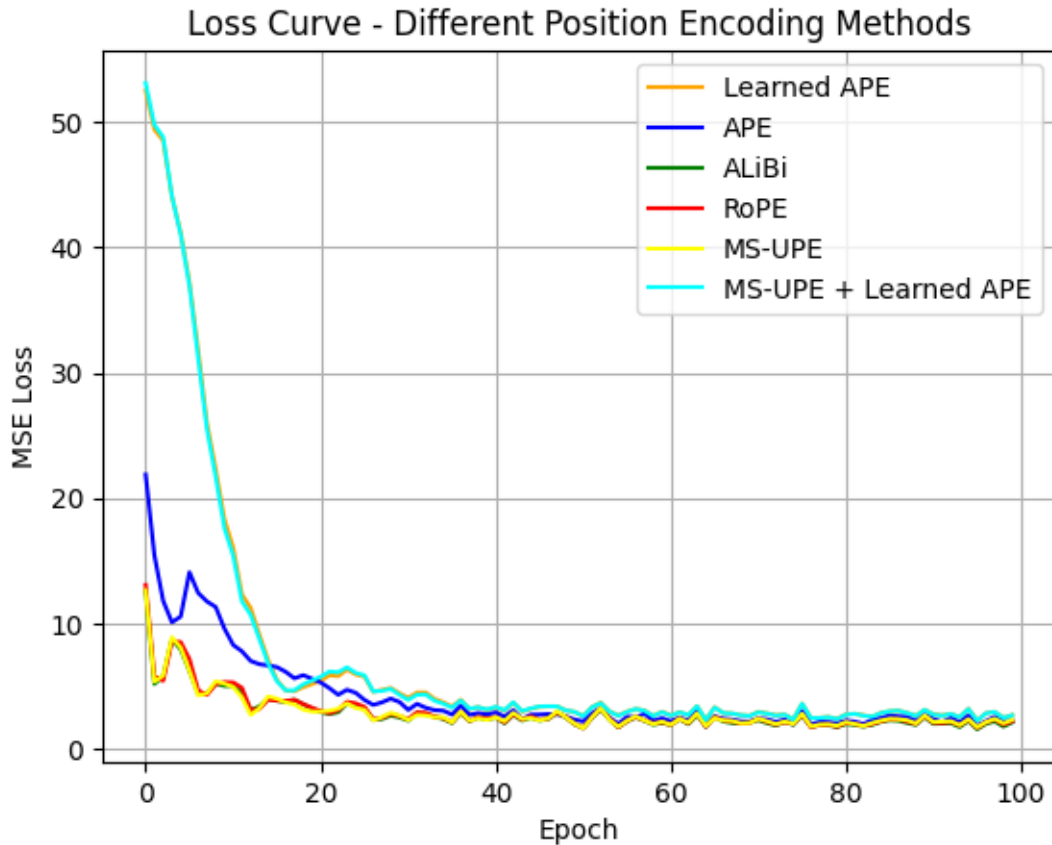
```



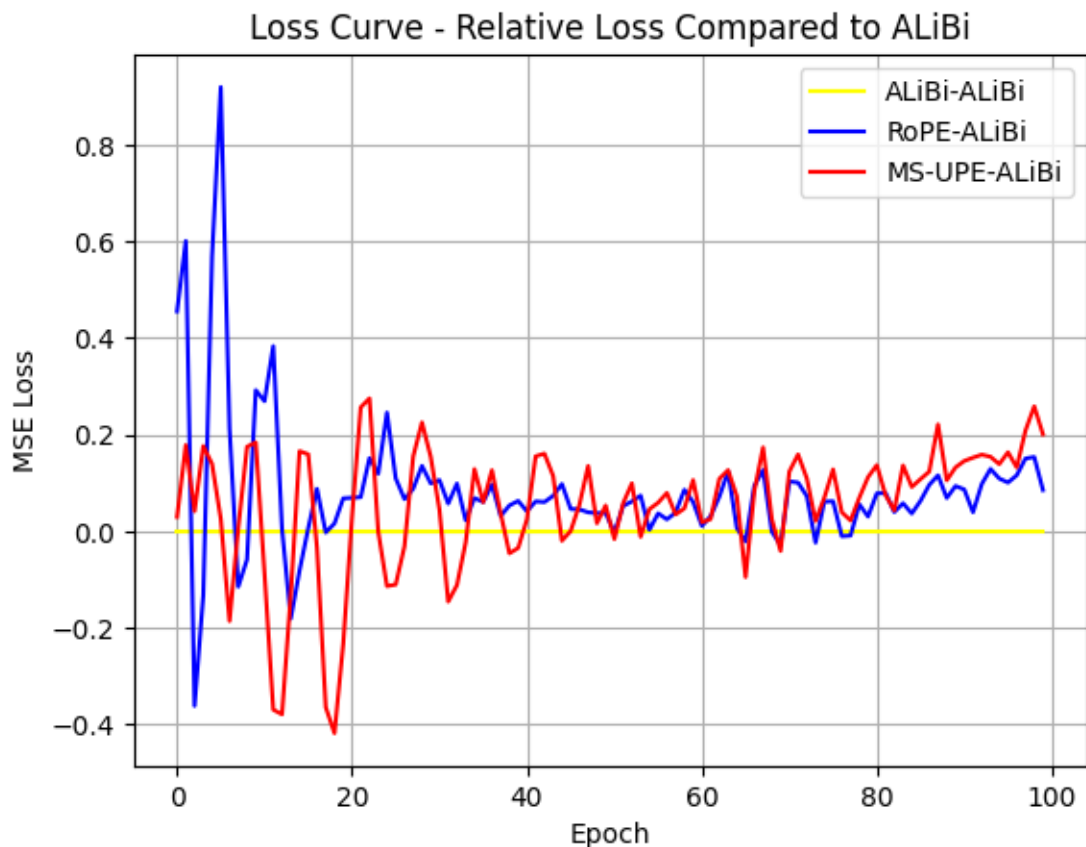
```
Epoch 45/100, Loss: 3.2357, current_blocks: 20
Epoch 50/100, Loss: 2.9586, current_blocks: 20
Epoch 55/100, Loss: 2.6367, current_blocks: 20
Epoch 60/100, Loss: 2.9363, current_blocks: 20
Epoch 65/100, Loss: 2.3051, current_blocks: 20
Epoch 70/100, Loss: 2.6297, current_blocks: 20
Epoch 75/100, Loss: 2.3529, current_blocks: 20
Epoch 80/100, Loss: 2.3792, current_blocks: 20
Epoch 85/100, Loss: 2.8603, current_blocks: 20
Epoch 90/100, Loss: 3.1174, current_blocks: 20
Epoch 95/100, Loss: 3.1854, current_blocks: 20
Epoch 100/100, Loss: 2.6862, current_blocks: 20
Training completed in 41.03 seconds.
```

画图

```
[92]: plt.plot(learning_ape_exp.loss_history, label='Learned APE', color='orange')
plt.plot(ape_exp.loss_history, label='APE', color='blue')
plt.plot(alibi_exp.loss_history, label='ALiBi', color='green')
plt.plot(rope_exp.loss_history, label='RoPE', color='red')
plt.plot(ms_upe_exp.loss_history, label='MS-UPE', color='yellow')
plt.plot(ms_upe_learned_ape_exp.loss_history, label='MS-UPE + Learned APE',
        color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Different Position Encoding Methods')
plt.grid()
plt.legend()
plt.show()
```



```
[ ]: plt.plot(list(map(sub, alibi_exp.loss_history, alibi_exp.loss_history)),  
             ↪label='ALiBi-ALiBi', color='yellow')  
plt.plot(list(map(sub, rope_exp.loss_history, alibi_exp.loss_history)),  
         ↪label='RoPE-ALiBi', color='blue')  
plt.plot(list(map(sub, ms_upe_exp.loss_history, alibi_exp.loss_history)),  
         ↪label='MS-UPE-ALiBi', color='red')  
plt.xlabel('Epoch')  
plt.ylabel('MSE Loss')  
plt.title('Loss Curve - Relative Loss Compared to ALiBi')  
plt.grid()  
plt.legend()  
plt.show()
```



sink padding 的有无对比实验

初始参数

```
[ ]: init_parameters, train_parameters = default_setup()
```

各个实验

```
[ ]: init_parameters['experiment_name'] = 'exp_without_sink_padding'
     experiment = LoopedTransformerExperiment(**init_parameters)
     experiment.train(**train_parameters)
```

exp_without_sink_padding initialized. Using device: mps

Epoch 10/100, Loss: 28.5238, current_blocks: 20

Epoch 20/100, Loss: 8.0300, current_blocks: 20

Epoch 30/100, Loss: 7.5267, current_blocks: 20

Epoch 40/100, Loss: 5.7270, current_blocks: 20

Epoch 50/100, Loss: 6.4822, current_blocks: 20

Epoch 60/100, Loss: 5.4732, current_blocks: 20
Epoch 70/100, Loss: 5.6123, current_blocks: 20
Epoch 80/100, Loss: 5.7751, current_blocks: 20
Epoch 90/100, Loss: 5.3166, current_blocks: 20
Epoch 100/100, Loss: 5.1050, current_blocks: 20
Training completed in 35.23 seconds.
exp_with_sink_padding_1 initialized. Using device: mps
Epoch 10/100, Loss: 29.0763, current_blocks: 20
Epoch 20/100, Loss: 8.1317, current_blocks: 20
Epoch 30/100, Loss: 7.6203, current_blocks: 20
Epoch 40/100, Loss: 5.7289, current_blocks: 20
Epoch 50/100, Loss: 6.5180, current_blocks: 20
Epoch 60/100, Loss: 5.4713, current_blocks: 20
Epoch 70/100, Loss: 5.6689, current_blocks: 20
Epoch 80/100, Loss: 5.7402, current_blocks: 20
Epoch 90/100, Loss: 5.3937, current_blocks: 20
Epoch 100/100, Loss: 5.2331, current_blocks: 20
Training completed in 35.00 seconds.
exp_with_sink_padding_2 initialized. Using device: mps
Epoch 10/100, Loss: 29.1206, current_blocks: 20
Epoch 20/100, Loss: 8.1602, current_blocks: 20
Epoch 30/100, Loss: 7.5729, current_blocks: 20
Epoch 40/100, Loss: 5.9227, current_blocks: 20
Epoch 50/100, Loss: 6.3734, current_blocks: 20
Epoch 60/100, Loss: 5.4039, current_blocks: 20
Epoch 70/100, Loss: 5.6607, current_blocks: 20
Epoch 80/100, Loss: 5.8500, current_blocks: 20
Epoch 90/100, Loss: 5.3117, current_blocks: 20
Epoch 100/100, Loss: 5.1204, current_blocks: 20
Training completed in 36.31 seconds.
exp_with_sink_padding_5 initialized. Using device: mps
Epoch 10/100, Loss: 26.2580, current_blocks: 20
Epoch 20/100, Loss: 8.0548, current_blocks: 20
Epoch 30/100, Loss: 7.3878, current_blocks: 20
Epoch 40/100, Loss: 5.6904, current_blocks: 20
Epoch 50/100, Loss: 6.2476, current_blocks: 20
Epoch 60/100, Loss: 5.4237, current_blocks: 20

Epoch 70/100, Loss: 5.6765, current_blocks: 20
Epoch 80/100, Loss: 5.8067, current_blocks: 20
Epoch 90/100, Loss: 5.1690, current_blocks: 20
Epoch 100/100, Loss: 5.0268, current_blocks: 20
Training completed in 40.53 seconds.

```
[ ]: init_parameters['experiment_name'] = 'exp_with_sink_padding_1'
train_parameters['sink_padding'] = 1
experiment_with_sink_padding_1 = LoopedTransformerExperiment(**init_parameters)
experiment_with_sink_padding_1.train(**train_parameters)
```

```
[ ]: init_parameters['experiment_name'] = 'exp_with_sink_padding_2'
train_parameters['sink_padding'] = 2
experiment_with_sink_padding_2 = LoopedTransformerExperiment(**init_parameters)
experiment_with_sink_padding_2.train(**train_parameters)
```

```
[ ]: init_parameters['experiment_name'] = 'exp_with_sink_padding_5'
train_parameters['sink_padding'] = 5
experiment_with_sink_padding_5 = LoopedTransformerExperiment(**init_parameters)
experiment_with_sink_padding_5.train(**train_parameters)
```

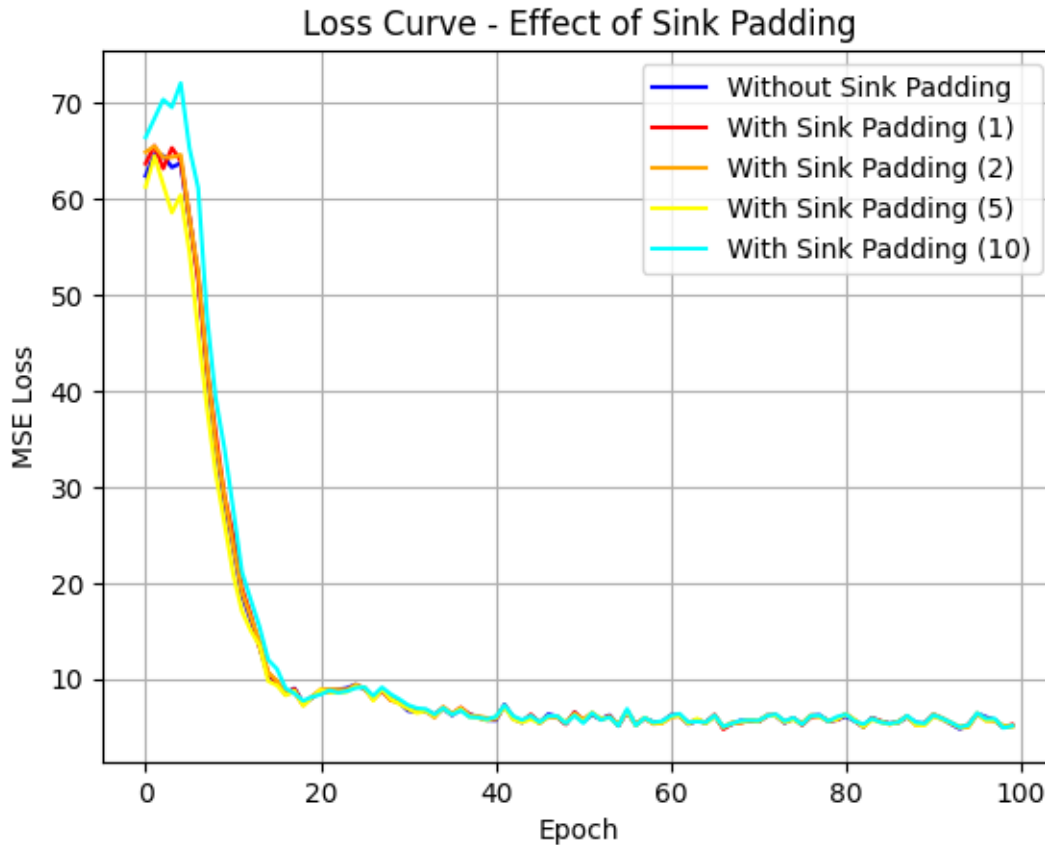
```
[138]: init_parameters['experiment_name'] = 'exp_with_sink_padding_10'
train_parameters['sink_padding'] = 10
experiment_with_sink_padding_10 =
↳ LoopedTransformerExperiment(**{**init_parameters, 'experiment_name':
↳ 'exp_with_sink_padding_10'})
experiment_with_sink_padding_10.train(**train_parameters)
```

exp_with_sink_padding_10 initialized. Using device: mps
Epoch 10/100, Loss: 34.0446, current_blocks: 20
Epoch 20/100, Loss: 8.1302, current_blocks: 20
Epoch 30/100, Loss: 7.8821, current_blocks: 20
Epoch 40/100, Loss: 5.8018, current_blocks: 20
Epoch 50/100, Loss: 6.1844, current_blocks: 20
Epoch 60/100, Loss: 5.6055, current_blocks: 20
Epoch 70/100, Loss: 5.6827, current_blocks: 20
Epoch 80/100, Loss: 6.0619, current_blocks: 20
Epoch 90/100, Loss: 5.3744, current_blocks: 20

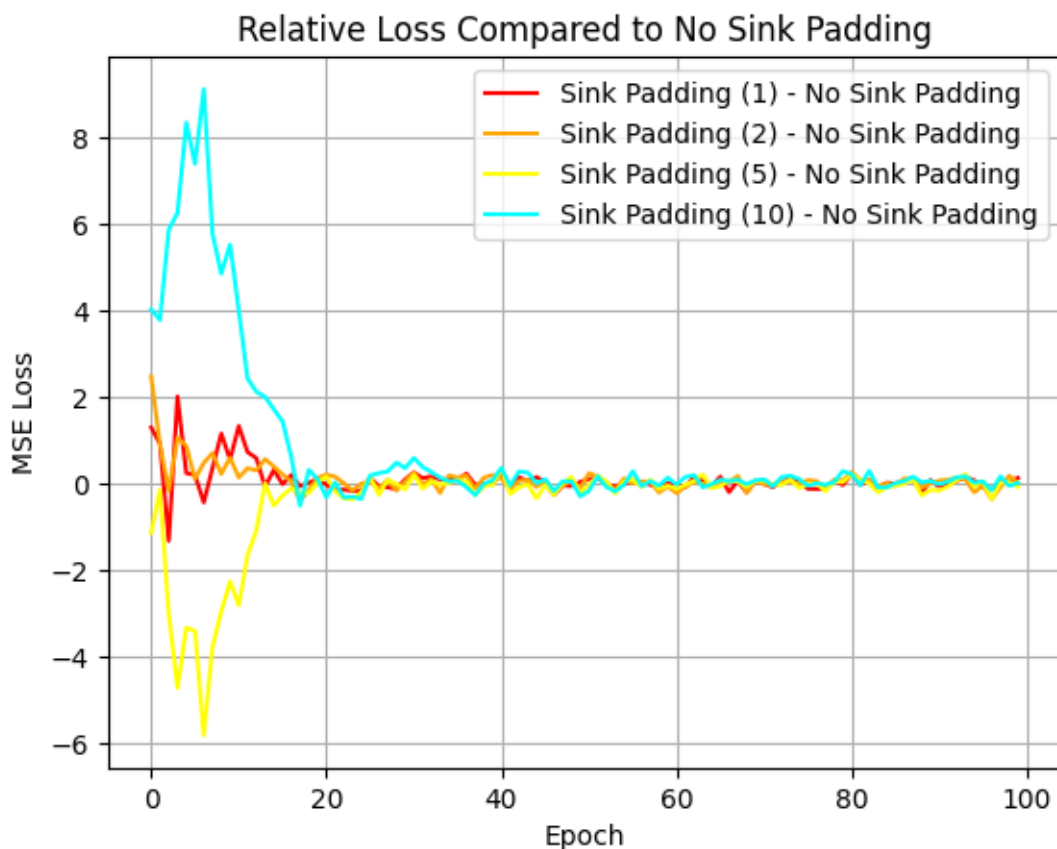
Epoch 100/100, Loss: 5.1151, current_blocks: 20
Training completed in 51.90 seconds.

画图

```
[ ]: plt.plot(experiment.loss_history, label='Without Sink Padding', color='blue')
plt.plot(experiment_with_sink_padding_1.loss_history, label='With Sink Padding_
↪(1)', color='red')
plt.plot(experiment_with_sink_padding_2.loss_history, label='With Sink Padding_
↪(2)', color='orange')
plt.plot(experiment_with_sink_padding_5.loss_history, label='With Sink Padding_
↪(5)', color='yellow')
plt.plot(experiment_with_sink_padding_10.loss_history, label='With Sink Padding_
↪(10)', color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Effect of Sink Padding')
plt.grid()
plt.legend()
plt.show()
```



```
[ ]: plt.plot(list(map(sub, experiment_with_sink_padding_1.loss_history, experiment.
    ↪loss_history)), label='Sink Padding (1) - No Sink Padding', color='red')
plt.plot(list(map(sub, experiment_with_sink_padding_2.loss_history, experiment.
    ↪loss_history)), label='Sink Padding (2) - No Sink Padding', color='orange')
plt.plot(list(map(sub, experiment_with_sink_padding_5.loss_history, experiment.
    ↪loss_history)), label='Sink Padding (5) - No Sink Padding', color='yellow')
plt.plot(list(map(sub, experiment_with_sink_padding_10.loss_history, experiment.
    ↪loss_history)), label='Sink Padding (10) - No Sink Padding', color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Relative Loss Compared to No Sink Padding')
plt.grid()
plt.legend()
plt.show()
```



Residual Gate 对比实验

初始参数

```
[45]: init_parameters, train_parameters = default_setup()
init_parameters['gate_lr_ratio'] = 100
train_parameters['scheduler_type'] = 'cosine'
train_parameters['epochs'] = 50
```

各个实验

```
[21]: init_parameters['experiment_name'] = 'exp_without_residual_gate'
experiment_no_residual_gate = LoopedTransformerExperiment(**init_parameters)
experiment_no_residual_gate.train(**train_parameters)
```

exp_without_residual_gate initialized in 0.52 seconds. Using device: mps

Epoch 5/50, Loss: 72.6460, current_blocks: 19

Epoch 10/50, Loss: 36.3084, current_blocks: 20


```
Epoch 15/50, Loss: 26.7078, current_blocks: 20
Epoch 20/50, Loss: 24.1986, current_blocks: 20
Epoch 25/50, Loss: 21.3926, current_blocks: 20
Epoch 30/50, Loss: 20.6963, current_blocks: 20
Epoch 35/50, Loss: 21.4774, current_blocks: 20
Epoch 40/50, Loss: 19.1252, current_blocks: 20
Epoch 45/50, Loss: 21.1858, current_blocks: 20
Epoch 50/50, Loss: 20.9827, current_blocks: 20
Training completed in 16.79 seconds.
```

```
[46]: init_parameters['experiment_name'] = 'exp_with_residual_gate_(1, 0.5)'
init_parameters['residual_gate'] = (1, 0.5)
init_parameters['residual_gate_type'] = 'learnable_scalar'
experiment_with_residual_gate_1_05 = LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_1_05.train(**train_parameters)
print(f"Learned residual gate values (1, 0.5) experiment: {experiment_with_residual_gate_1_05.residual_gate_values}")
```

```
exp_with_residual_gate_(1, 0.5) initialized in 0.02 seconds. Using device: mps
Epoch 5/50, Loss: 24.6447, current_blocks: 19
Epoch 10/50, Loss: 19.1647, current_blocks: 20
Epoch 15/50, Loss: 22.0188, current_blocks: 20
Epoch 20/50, Loss: 21.6364, current_blocks: 20
Epoch 25/50, Loss: 19.9177, current_blocks: 20
Epoch 30/50, Loss: 19.2544, current_blocks: 20
Epoch 35/50, Loss: 20.4781, current_blocks: 20
Epoch 40/50, Loss: 18.1403, current_blocks: 20
Epoch 45/50, Loss: 20.3567, current_blocks: 20
Epoch 50/50, Loss: 20.1678, current_blocks: 20
Training completed in 17.17 seconds.
Learned residual gate values (1, 0.5) experiment: [0.8780055 0.4036405]
```

```
[47]: init_parameters['experiment_name'] = 'exp_with_residual_gate_(1, 0.5)'
init_parameters['residual_gate'] = (1, 0.5)
init_parameters['residual_gate_type'] = 'learnable_vector'
```

```

experiment_with_residual_gate_1_05_vector =
    ↳ LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_1_05_vector.train(**train_parameters)
print(f"Learned residual gate values (1, 0.5) experiment:
    ↳ mean={experiment_with_residual_gate_1_05_vector.residual_gate_values.
    ↳ mean(axis=0)}, std={experiment_with_residual_gate_1_05_vector.
    ↳ residual_gate_values.std(axis=0)}")

```

exp_with_residual_gate_(1, 0.5) initialized in 0.01 seconds. Using device: mps

Epoch 5/50, Loss: 23.3988, current_blocks: 19

Epoch 10/50, Loss: 20.6244, current_blocks: 20

Epoch 15/50, Loss: 21.7083, current_blocks: 20

Epoch 20/50, Loss: 21.5294, current_blocks: 20

Epoch 25/50, Loss: 20.3255, current_blocks: 20

Epoch 30/50, Loss: 19.2366, current_blocks: 20

Epoch 35/50, Loss: 20.7056, current_blocks: 20

Epoch 40/50, Loss: 18.3417, current_blocks: 20

Epoch 45/50, Loss: 20.6420, current_blocks: 20

Epoch 50/50, Loss: 20.4405, current_blocks: 20

Training completed in 17.71 seconds.

Learned residual gate values (1, 0.5) experiment: mean=[0.95619005 0.47291675],
std=[0.03953942 0.04971235]

```

[49]: init_parameters['experiment_name'] = 'exp_with_residual_gate_(1, 0)'
init_parameters['residual_gate'] = (1, 0)
init_parameters['residual_gate_type'] = 'learnable_scalar'
experiment_with_residual_gate_1_0 =
    ↳ LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_1_0.train(**train_parameters)
print(f"Learned residual gate values (1, 0) experiment:
    ↳ {experiment_with_residual_gate_1_0.residual_gate_values}")

```

exp_with_residual_gate_(1, 0) initialized in 0.02 seconds. Using device: mps

Epoch 5/50, Loss: 22.2907, current_blocks: 19

Epoch 10/50, Loss: 19.2169, current_blocks: 20

Epoch 15/50, Loss: 21.2251, current_blocks: 20

Epoch 20/50, Loss: 21.2406, current_blocks: 20

```
Epoch 25/50, Loss: 19.9645, current_blocks: 20
Epoch 30/50, Loss: 19.1195, current_blocks: 20
Epoch 35/50, Loss: 20.4177, current_blocks: 20
Epoch 40/50, Loss: 18.0073, current_blocks: 20
Epoch 45/50, Loss: 20.3680, current_blocks: 20
Epoch 50/50, Loss: 20.0981, current_blocks: 20
Training completed in 17.11 seconds.
Learned residual gate values (1, 0) experiment: [0.86226875 0.07368369]
```

```
[50]: init_parameters['experiment_name'] = 'exp_with_residual_gate_(1, 0)_vector'
init_parameters['residual_gate'] = (1, 0)
init_parameters['residual_gate_type'] = 'learnable_vector'
experiment_with_residual_gate_1_0_vector = □
↳ LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_1_0_vector.train(**train_parameters)
print(f"Learned residual gate values (1, 0) experiment: □
↳ mean={experiment_with_residual_gate_1_0_vector.residual_gate_values.
↳ mean(axis=0)}, std={experiment_with_residual_gate_1_0_vector.
↳ residual_gate_values.std(axis=0)}")
```

```
exp_with_residual_gate_(1, 0)_vector initialized in 0.01 seconds. Using device:
mps
```

```
Epoch 5/50, Loss: 23.8717, current_blocks: 19
Epoch 10/50, Loss: 19.8342, current_blocks: 20
Epoch 15/50, Loss: 21.6780, current_blocks: 20
Epoch 20/50, Loss: 21.1577, current_blocks: 20
Epoch 25/50, Loss: 20.1856, current_blocks: 20
Epoch 30/50, Loss: 19.0574, current_blocks: 20
Epoch 35/50, Loss: 20.4849, current_blocks: 20
Epoch 40/50, Loss: 18.0559, current_blocks: 20
Epoch 45/50, Loss: 20.5518, current_blocks: 20
Epoch 50/50, Loss: 20.1890, current_blocks: 20
Training completed in 17.75 seconds.
Learned residual gate values (1, 0) experiment: mean=[0.97792023 0.01803507],
std=[0.0346687 0.03598361]
```

```
[51]: init_parameters['experiment_name'] =_
      ↪ 'exp_with_residual_gate_random_learnable_scalar'
init_parameters['residual_gate'] = 'random'
init_parameters['residual_gate_type'] = 'learnable_scalar'
init_parameters['residual_random'] = (0.9, 0.1)
experiment_with_residual_gate_learnable_scalar =_
      ↪ LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_learnable_scalar.train(**train_parameters)
print(f"Learnable Scalar Residual Gate Values:_
      ↪ {experiment_with_residual_gate_learnable_scalar.residual_gate_values}")
```

exp_with_residual_gate_random_learnable_scalar initialized in 0.01 seconds.

Using device: mps

Epoch 5/50, Loss: 20.8480, current_blocks: 19
 Epoch 10/50, Loss: 18.6310, current_blocks: 20
 Epoch 15/50, Loss: 21.2739, current_blocks: 20
 Epoch 20/50, Loss: 21.1260, current_blocks: 20
 Epoch 25/50, Loss: 19.8267, current_blocks: 20
 Epoch 30/50, Loss: 19.0972, current_blocks: 20
 Epoch 35/50, Loss: 20.3443, current_blocks: 20
 Epoch 40/50, Loss: 18.0375, current_blocks: 20
 Epoch 45/50, Loss: 20.3402, current_blocks: 20
 Epoch 50/50, Loss: 20.1229, current_blocks: 20

Training completed in 17.09 seconds.

Learnable Scalar Residual Gate Values: [0.49568892 0.7366962]

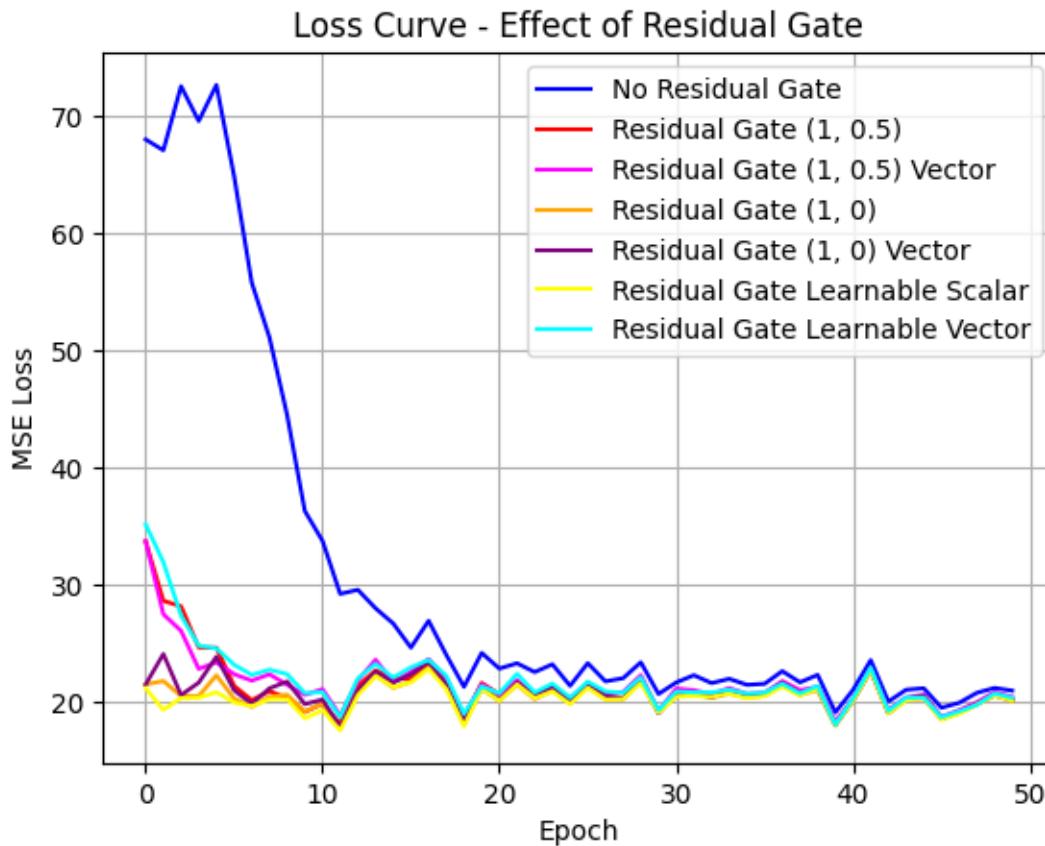
```
[52]: init_parameters['experiment_name'] = 'exp_with_residual_gate_learnable_vector'
init_parameters['residual_gate'] = 'random'
init_parameters['residual_gate_type'] = 'learnable_vector'
init_parameters['residual_random'] = (0.9, 0.1)
experiment_with_residual_gate_learnable_vector =_
      ↪ LoopedTransformerExperiment(**init_parameters)
experiment_with_residual_gate_learnable_vector.train(**train_parameters)
print(f"Learnable Vector Residual Gate Values:_
      ↪ mean={experiment_with_residual_gate_learnable_vector.residual_gate_values.
      ↪ mean(axis=0)}, std={experiment_with_residual_gate_learnable_vector.
      ↪ residual_gate_values.std(axis=0)}")
```

```
exp_with_residual_gate_learnable_vector initialized in 0.01 seconds. Using
device: mps
Epoch 5/50, Loss: 24.5847, current_blocks: 19
Epoch 10/50, Loss: 20.7673, current_blocks: 20
Epoch 15/50, Loss: 22.1009, current_blocks: 20
Epoch 20/50, Loss: 21.3415, current_blocks: 20
Epoch 25/50, Loss: 20.3913, current_blocks: 20
Epoch 30/50, Loss: 19.3258, current_blocks: 20
Epoch 35/50, Loss: 20.7225, current_blocks: 20
Epoch 40/50, Loss: 18.0786, current_blocks: 20
Epoch 45/50, Loss: 20.3941, current_blocks: 20
Epoch 50/50, Loss: 20.3768, current_blocks: 20
Training completed in 17.60 seconds.
Learnable Vector Residual Gate Values: mean=[0.8660181  0.86898607],
std=[0.10505898 0.11353411]
```

画图

```
[53]: plt.plot(experiment_no_residual_gate.loss_history, label='No Residual Gate',
             color='blue')
plt.plot(experiment_with_residual_gate_1_05.loss_history, label='Residual Gate_
             (1, 0.5)', color='red')
plt.plot(experiment_with_residual_gate_1_05_vector.loss_history,
             label='Residual Gate (1, 0.5) Vector', color='magenta')
plt.plot(experiment_with_residual_gate_1_0.loss_history, label='Residual Gate_
             (1, 0)', color='orange')
plt.plot(experiment_with_residual_gate_1_0_vector.loss_history, label='Residual_
             Gate (1, 0) Vector', color='purple')
plt.plot(experiment_with_residual_gate_learnable_scalar.loss_history,
             label='Residual Gate Learnable Scalar', color='yellow')
plt.plot(experiment_with_residual_gate_learnable_vector.loss_history,
             label='Residual Gate Learnable Vector', color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Effect of Residual Gate')
plt.grid()
plt.legend()
```

```
plt.show()
```

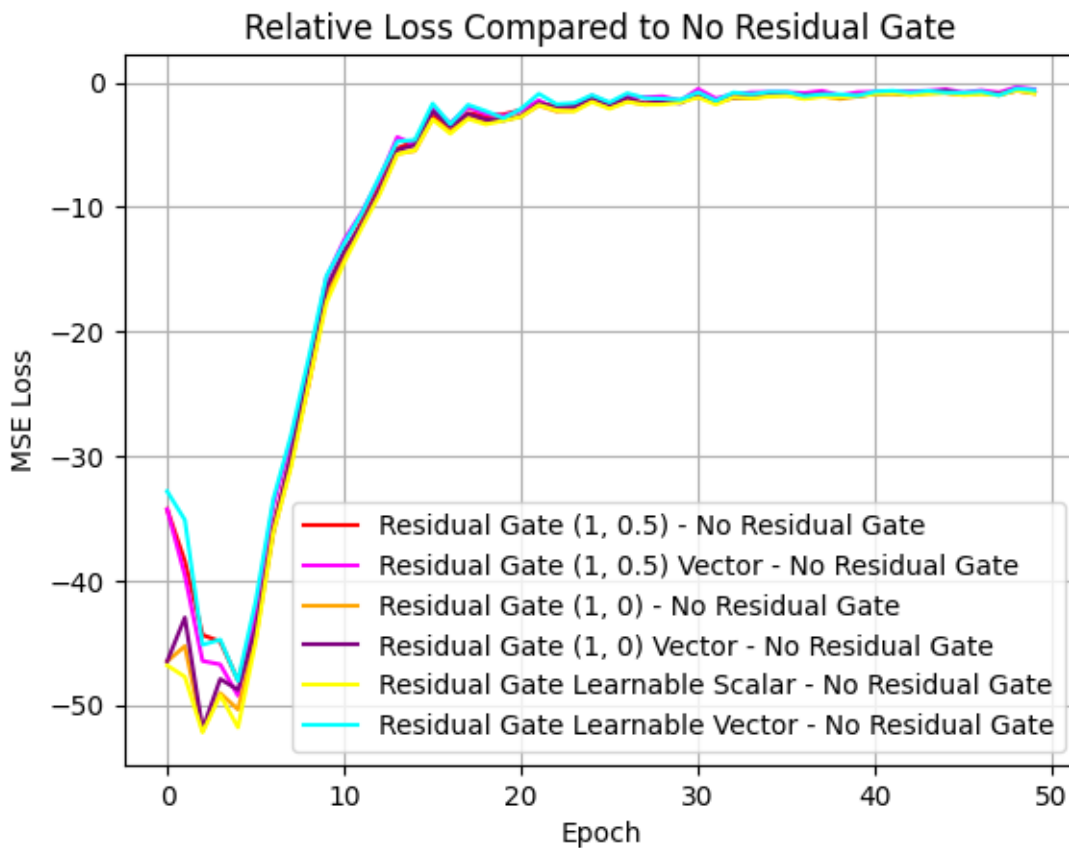


```
[56]: plt.plot(list(map(sub, experiment_with_residual_gate_1_05.loss_history,
    ↪experiment_no_residual_gate.loss_history)), label='Residual Gate (1, 0.5) -
    ↪No Residual Gate', color='red')
plt.plot(list(map(sub, experiment_with_residual_gate_1_05_vector.loss_history,
    ↪experiment_no_residual_gate.loss_history)), label='Residual Gate (1, 0.5)
    ↪Vector - No Residual Gate', color='magenta')
plt.plot(list(map(sub, experiment_with_residual_gate_1_0.loss_history,
    ↪experiment_no_residual_gate.loss_history)), label='Residual Gate (1, 0) - No
    ↪Residual Gate', color='orange')
plt.plot(list(map(sub, experiment_with_residual_gate_1_0_vector.loss_history,
    ↪experiment_no_residual_gate.loss_history)), label='Residual Gate (1, 0)
    ↪Vector - No Residual Gate', color='purple')
```

```

plt.plot(list(map(sub, experiment_with_residual_gate_learnable_scalar.
    ↪loss_history, experiment_no_residual_gate.loss_history)), label='Residual_
    ↪Gate Learnable Scalar - No Residual Gate', color='yellow')
plt.plot(list(map(sub, experiment_with_residual_gate_learnable_vector.
    ↪loss_history, experiment_no_residual_gate.loss_history)), label='Residual_
    ↪Gate Learnable Vector - No Residual Gate', color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Relative Loss Compared to No Residual Gate')
plt.grid()
plt.legend()
plt.show()

```



```
[60]: # 把 loss 和 residual 的变化画在一张图上
# 在一张图中呈现四张子图，分别代表 learnable_scalar 的 (1,0.5) 和 random 实验，以及
# learnable_vector 的 (1,0.5) 和 random 实验，每张子图中同时展示 loss 曲线和
# residual_gate_history 曲线，方便比较它们之间的关系
fig, axs = plt.subplots(2, 2, figsize=(15, 10))
# (1, 0.5) learnable scalar
axs[0, 0].plot(experiment_with_residual_gate_1_05.loss_history, label='Loss',
               ↪color='blue')
axs[0, 0].set_xlabel('Epoch')
axs[0, 0].set_ylabel('MSE Loss', color='blue')
axs[0, 0].tick_params(axis='y', labelcolor='blue')
ax2_1 = axs[0, 0].twinx()
ax2_1_history = experiment_with_residual_gate_1_05.residual_gate_history
ax2_1.plot([(h - ax2_1_history[0])[0] for h in ax2_1_history], label='Residual_
↪Gate a relative', color='red')
ax2_1.plot([(h - ax2_1_history[0])[1] for h in ax2_1_history], label='Residual_
↪Gate b relative', color='orange')
ax2_1.set_ylabel('Residual Gate Values relative', color='red')
ax2_1.tick_params(axis='y', labelcolor='red')
axs[0, 0].set_title('Residual Gate (1, 0.5) Learnable Scalar relative to_
↪Initial Value')
axs[0, 0].legend(loc='upper left')
ax2_1.legend(loc='upper right')
# (1, 0.5) learnable vector
axs[0, 1].plot(experiment_with_residual_gate_1_05_vector.loss_history,
               ↪label='Loss', color='blue')
axs[0, 1].set_xlabel('Epoch')
axs[0, 1].set_ylabel('MSE Loss', color='blue')
axs[0, 1].tick_params(axis='y', labelcolor='blue')
ax2_2 = axs[0, 1].twinx()
ax2_history = experiment_with_residual_gate_1_05_vector.residual_gate_history
ax2_2.plot([ax2_history[i][:,0].mean() - ax2_history[0][:,0].mean() for i in_
↪range(len(ax2_history))], label='Residual Gate a mean relative', color='red')
ax2_2.plot([ax2_history[i][:,1].mean() - ax2_history[0][:,1].mean() for i in_
↪range(len(ax2_history))], label='Residual Gate b mean relative',
               ↪color='orange')
```



```

ax2_2.set_ylabel('Residual Gate Mean Values relative', color='red')
ax2_2.tick_params(axis='y', labelcolor='red')
axs[0, 1].set_title('Residual Gate (1, 0.5) Learnable Vector relative to_
↳Initial Value')
axs[0, 1].legend(loc='upper left')
ax2_2.legend(loc='upper right')
# random learnable scalar
axs[1, 0].plot(experiment_with_residual_gate_learnable_scalar.loss_history,
↳label='Loss', color='blue')
axs[1, 0].set_xlabel('Epoch')
axs[1, 0].set_ylabel('MSE Loss', color='blue')
axs[1, 0].tick_params(axis='y', labelcolor='blue')
ax2_3 = axs[1, 0].twinx()
ax2_3.plot([h[0] - experiment_with_residual_gate_learnable_scalar.
↳residual_gate_history[0][0] for h in_
↳experiment_with_residual_gate_learnable_scalar.residual_gate_history],
↳label='Residual Gate a relative', color='red')
ax2_3.plot([h[1] - experiment_with_residual_gate_learnable_scalar.
↳residual_gate_history[0][1] for h in_
↳experiment_with_residual_gate_learnable_scalar.residual_gate_history],
↳label='Residual Gate b relative', color='orange')
ax2_3.set_ylabel('Residual Gate Values relative', color='red')
ax2_3.tick_params(axis='y', labelcolor='red')
axs[1, 0].set_title('Residual Gate Random Learnable Scalar relative to Initial_
↳Value')
axs[1, 0].legend(loc='upper left')
ax2_3.legend(loc='upper right')
# random learnable vector
axs[1, 1].plot(experiment_with_residual_gate_learnable_vector.loss_history,
↳label='Loss', color='blue')
axs[1, 1].set_xlabel('Epoch')
axs[1, 1].set_ylabel('MSE Loss', color='blue')
axs[1, 1].tick_params(axis='y', labelcolor='blue')
ax2_4 = axs[1, 1].twinx()
ax2_history = experiment_with_residual_gate_learnable_vector.
↳residual_gate_history

```

```

ax2_4.plot([ax2_history[i][:,0].mean() - ax2_history[0][:,0].mean() for i in
    range(len(ax2_history))], label='Residual Gate a mean relative', color='red')
ax2_4.plot([ax2_history[i][:,1].mean() - ax2_history[0][:,1].mean() for i in
    range(len(ax2_history))], label='Residual Gate b mean relative',
    color='orange')
ax2_4.set_ylabel('Residual Gate Mean Values relative', color='red')
ax2_4.tick_params(axis='y', labelcolor='red')
axs[1, 1].set_title('Residual Gate Random Learnable Vector relative to Initial
    Value')
axs[1, 1].legend(loc='upper left')
ax2_4.legend(loc='upper right')
plt.suptitle('Loss and Residual Gate Values Over Epochs')
plt.tight_layout()
plt.show()

```

